



NuWro Workshop, Wrocław, Poland

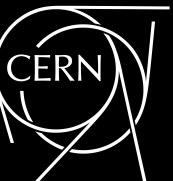
17th June 2026

NuBib and ADoNIS: New Tools for Neutrino Interaction Physics



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More experiments & more data...

How do we keep track of what has been done, what is missing, and the trends in the field?

A view from neutrino scattering community:

INSPIRE-HEP <http://inspirehep.net>

Good: Comprehensive literature DB. Linked references (arXiv, Journal) & authors. Continuously updated.

Missing: Some option that allows to identify/filter papers by topic (e.g. all $CC0\pi$ measurements). Datasets.

HEPData & Zenodo <https://www.hepdata.net/> <https://zenodo.org/>

Good: Provide data releases for some published measurements

Missing: A large fraction of old and new neutrino results. Depends on people willing to upload data in there.

NUISANCE <https://nuisance.hepforge.org/>

Good: Largest collection of neutrino data releases and measurements in one place + tools to actually use it.

Missing: Some neutrino results and a convenient interface to browse and filter existing data (a nice visual catalog).

Experiment	Measurement	Target(s)
ArgoNeuT	2p [327]	Ar
K2K	M_A [328]	H ₂ O
MicroBooNE	$\frac{d\sigma}{dQ^2}, \frac{d\sigma}{dp_{\mu,p}}, \frac{d\sigma}{d\cos\theta_{\mu,p}}$ [329]	Ar
MINER ν A	$\frac{d\sigma}{dQ^2}$ [330, 331, 332], 1p [333], ν_e [334] $\frac{d^2\sigma}{dp_T d_{ }}$ [335, 336], $\frac{d\sigma}{dp_n} \frac{d\sigma}{d\delta_{\alpha T}}$ [337], $\frac{d^2\sigma}{dE_{avail} dq_3}$ [338]	CH, Fe, Pb
MiniBooNE	$\frac{d^2\sigma}{dT_{\mu} d\theta_{\mu}}$ [339, 340], M_A [341], NC [342, 343]	CH ₂
MINOS	M_A [344]	Fe
NOMAD	$M_{A,\sigma}(E_{\nu})$ [345]	C
Super-K	NC [346]	H ₂ O
T2K	$\frac{d^2\sigma}{dT_{\mu} d\theta_{\mu}}$ [252, 347, 348, 349, 350], $\sigma(E_{\nu})$ [351], M_A [352] NC [353, 354], $\frac{d\sigma}{d\delta_{pT}} \frac{d\sigma}{d\delta_{\alpha T}}$ [355]	CH, H ₂ O

TABLE 3.4: Summary of quasielastic cross-section measurements by modern neutrino experiments. Adapted from Ref. [301].

Very painful, and can't be recasted!

Experiment	$\pi^{\pm}$ Measurement	π^0 Measurement	Target(s)
ArgoNeuT	CC [356]	NC [357]	Ar
K2K	CC [358, 359]	CC [360], NC [361]	H ₂ O
MicroBooNE	-	CC [362]	Ar
MINER ν A	CC [363, 364, 365, 366, 367]	CC [364, 368, 369], NC [370]	CH
MiniBooNE	CC [371, 372]	CC [373], NC [374, 375]	CH ₂
MINOS	-	NC [376]	Fe
NOMAD	-	NC [377]	C
NO ν A	-	NC [378]	C
SciBooNE	CC [379]	NC [380, 381]	CH
T2K	CC [382, 383, 384], NC [CJV]	NC [385]	CH ₂ , H ₂ O

TABLE 3.5: Summary of neutrino induced pion production cross-section measurements by modern neutrino experiments. The tag [CJV] highlights the space meant to be filled by the study later presented in this thesis. Table adapted from

A long time wish....

- It would be great to have a simple webpage that would allow me to better understand what has been done in the past, the trends in the field, spot measurements that would fill important gaps in the literature, etc.
- Over the years I learned from many people in the community they would also love having such a tool.

So, why did this not exist already?

Technical reasons

- Challenge 1 **[Indexing]**: Search the existing literature (hundreds of papers!), triage, and classify it with tags.
- Challenge 2 **[Visuals]**: Gather the indexed information into a nice webpage that can be organically extended.

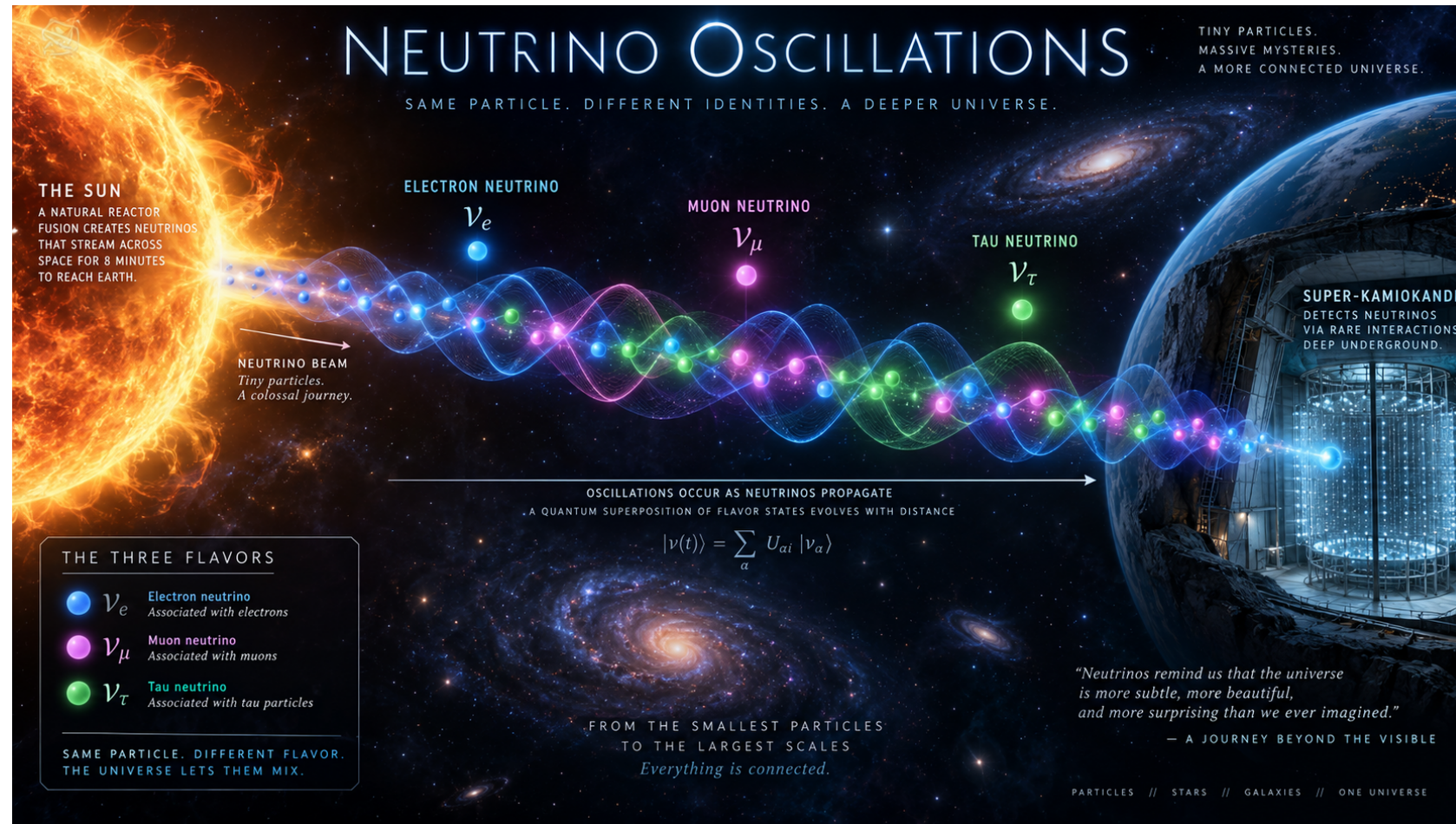
Practical reasons

- We don't have endless time, and we are not website designers...
- Incentives for an individual to take responsibility on this is questionable (system rewards "hardcore science")

However...

- In recent years Agentic AI tools have radically redefined the meaning of "challenging" across a wide range of tasks.

AI SLOP is real and should NOT be ignored



***AI is not and can be not an excuse
for bad science.***



***It is a responsibility of the whole
community to fight against bad
AI use.***

However it's usefulness cant be denied

<https://openai.com/index/navier-stokes-solution/>

FINITE TIME BLOWUP FOR NAVIER-STOKES

OPENAI

ABSTRACT. For every positive viscosity, we construct a solution of the three-dimensional incompressible Navier-Stokes equations that starts from rest and develops unbounded velocity in finite time while maintaining uniformly bounded kinetic energy.

<https://www.scientificamerican.com/article/openais-latest-math-breakthroughs-commit-research-misconduct-experts-say/>

***And it is already changing publishing
trends in the field***

<https://arxiv.org/abs/2607.10780>

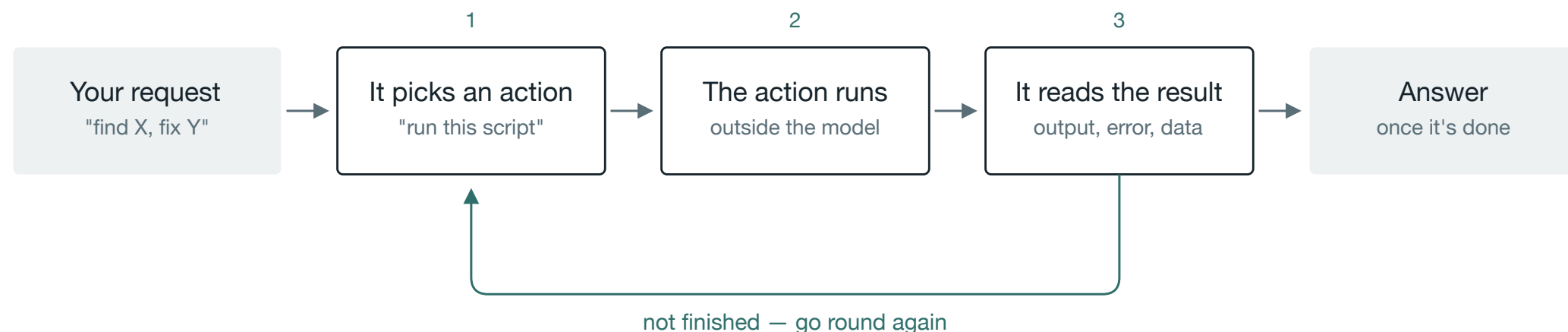
**So, how can we actually make a
good use of AI in science?**

A bit of historical context...

- I have been an extremely early user of Agentic AI tools (wrt to the neutrino physics community).
- I remember playing with AutoGPT in Apr 2023, and thinking “oh, this is the future!”.
- Ever since I have been progressively shifting my research workflows to be fully Agent driven.
- In Feb 2025 “Claude Code” was released, and has been every since the primary engine behind my research work.

A quick look into the concept

- Since these tools are fairly new, let me give you an overview of the concept:



- This simple concept allows **[IF USED WELL!]** to offload massively time consuming parts of any project.

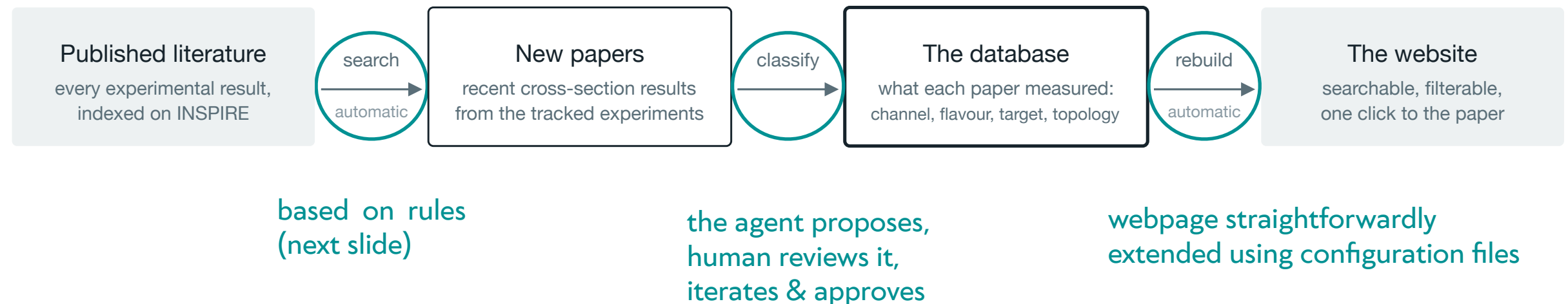
NuBIB

- In 2026, I realised coding webpages for science is very easy with this tools, and took inspiration from S. Young webpage for a different project I am working on https://www.deeplearnphysics.org/doraemon_site/

NuBib is, at its core, an “Agentic workflow”

- If you simply ask to an agent “classify existing neutrino interactions literature” outcome will be useless and unreliable.

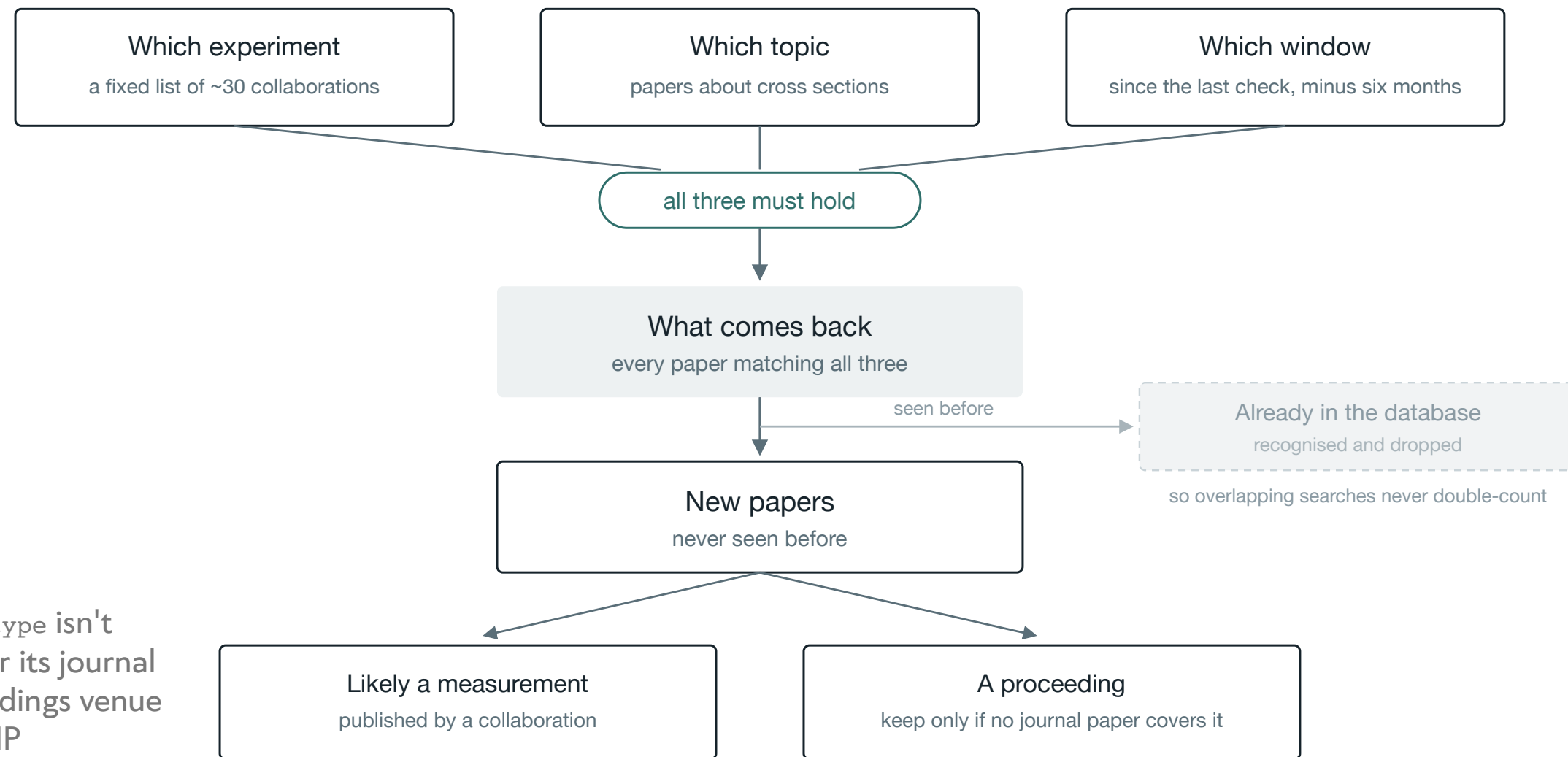
The idea is to enforce the same loop one would do “by hand” but having agents take care of the specifics steps that can be automated



“Utilities” are provided as python scripts that enforce some actions (e.g. do some queries in INSPIRE-HEP API)

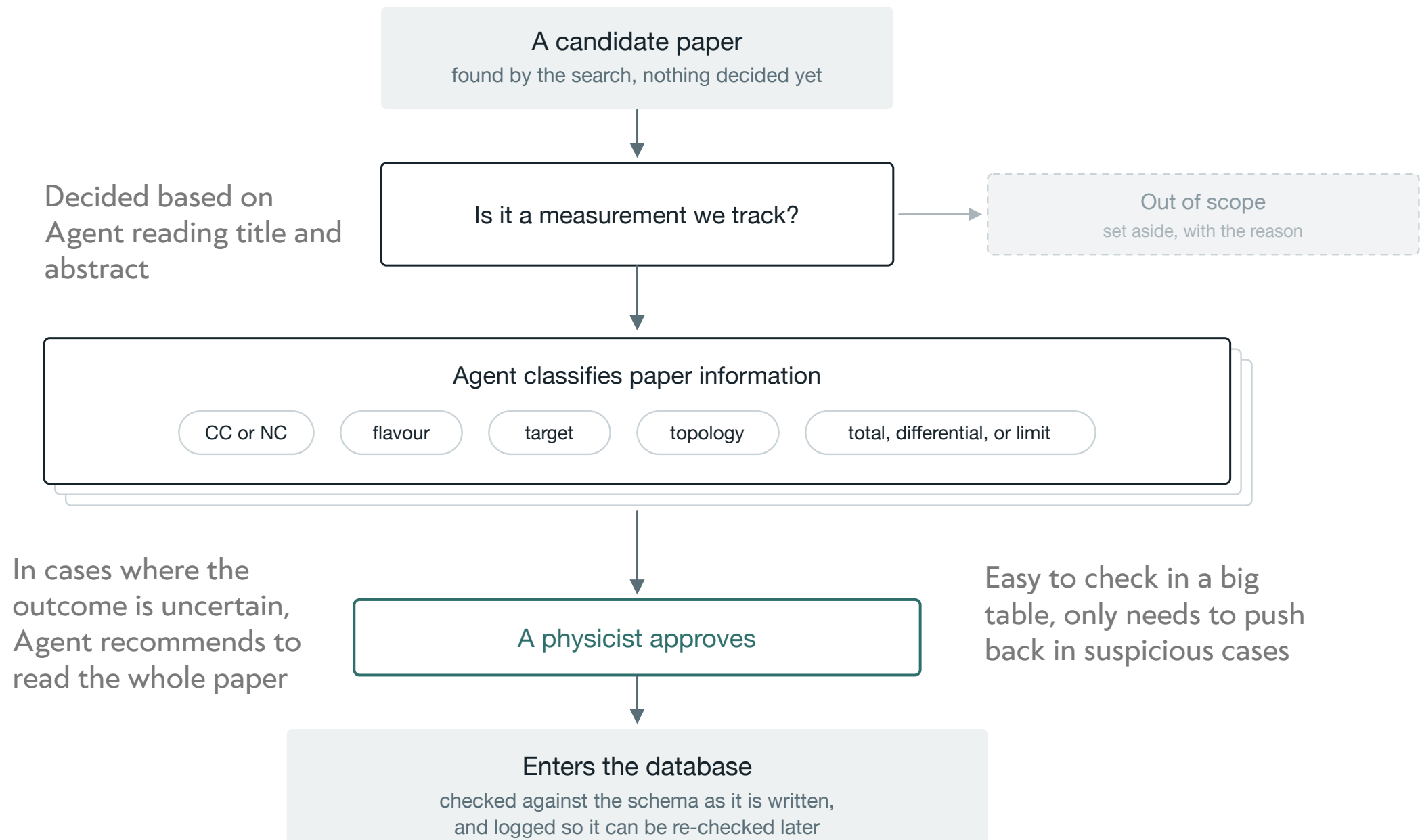
The Agent uses those utilities to execute actions with planned consistency.

For instance, the search engine uses (among others) https://github.com/cesarjesusvalls/NuIntBib/blob/main/scripts/find_new_papers.py



INSPIRE's
document_type isn't
"article" (or its journal
is a proceedings venue
like PoS/AIP
Conf.Proc)

The search output is preserved in a .yaml file and the program proceeds to classifying the candidate papers



- Some scripts take care of all the manual labor, e.g. formatting checks, linking papers to specific collaborations, etc

The webpage itself is re-written every time using the latest status of the database.

- Scripts use the config files that define each paper and automatically print a fresh web page for each, so the website is really just an always-up-to-date printout of the underlying database, with nothing on it written by hand.
- I personally check how new entries look like in the webpage locally before pushing to changes to nubib.org

The webpage itself is hosted using GitHub pages (free)

- Learn more about it here: <https://docs.github.com/en/pages>
- The domain (nubib.org) was acquired by ~8€/year.

Long term?

- It is difficult to get started because one needs to get “up to date”. However, reviewing new papers additions is very easy. Just need to ask to an Agent every once in a while “Check for updates” to initiate the process.

Minimal effort & cost to maintain!

NuBib
by César Jesús Valls

Neutrino measurements, in one place.

Papers Table builder Statistics About

NEUTRINO MEASUREMENTS LITERATURE RESOURCE

Neutrino measurements, in one place.

NuBib is a continuously updated index of published neutrino measurements, covering both interaction and oscillation results. Search and filter by experiment, channel, and more, with one-click links to arXiv, the journal, INSPIRE, and BibTeX.

Browse interaction papers →
222 tracked

Browse oscillation measurements →
204 tracked

What originally started as neutrino interactions
only now also supports oscillations!

BY EXPERIMENT

Coverage across the field

Spanning 1973 to 2026, the database tracks 426 interaction and oscillation papers across 50 experiments, with 106,448 INSPIRE citations in total.

54 T2K	44 MINERvA	34 MicroBooNE	29 Super-Kamiokande	24 MINOS	23 MiniBooNE	21 NOvA	15 IceCube	14 Daya Bay
13 OPERA	12 NOMAD	11 ANL	10 BNL	10 K2K	9 RENO	9 Double Chooz	8 SNO	7 COHERENT
7 ArgoNeuT	7 LSND	6 Gargamelle	6 SciBooNE	5 KM3NeT	4 BEBC	4 CHARM	4 STEREO	4 KamLAND
3 NINJA	3 SKAT	3 FNAL	3 ICARUS	3 PROSPECT	2 FASER	2 Aachen	2 NEOS	2 ANTARES
2 Borexino	1 SND@LHC	1 CONUS	1 Dresden-II	1 NuTeV	1 CHORUS	1 CHARM-II	1 CCFR	1 Serpukhov
1 MINOS+	1 NEUTRINO-4	1 DANSS	1 SAGE	1 GALLEX				

NuBib
by César Jesús Valls

Neutrino measurements, in one place.

[Papers](#)[Table builder](#)[Statistics](#)[About](#)

426 PAPERS

Neutrino measurements

Up-to-date, labeled neutrino interaction and oscillation papers.

Interactions 222

Oscillations 204

All experiments 222

Aachen 2

ANL 11

ArgoNeUT 7

BEBC 4

BNL 10

CCFR 1

CHARM 4

CHARM-II 1

CHORUS 1

COHERENT 7

CONUS 1

Dresden-II 1

FASER 2

FNAL 3

Gargamelle 6

ICARUS 1

K2K 5

MicroBooNE 30

MINERvA 44

Experiment

MiniBooNE 11

MINOS 3

NINJA 3

NOMAD 12

NOvA 8

NuTeV 1

SciBooNE 4

Serpukhov 1

SKAT 3

SND@LHC 1

Super-Kamiokande 2

T2K 32

Filter by

[Clear all](#)

Current

CC+NC 222

CC 177

NC 46

Flavor

All flavors 222

 ν_e 22 $\bar{\nu}_e$ 9 ν_μ 182 $\bar{\nu}_\mu$ 70

Target

All targets 222

CH 65

Ar 39

C 34

D₂ 16

Fe 20

CH₂ 16H₂ 11H₂O 13

Pb 10

CF₃Br 6C₃H₈ 5

O 6

CaCO₃ 5

Ge 4

Al 4

W 3

CsI 2

Rock 1

I 1

H 1

Cu 1

SiO₂ 1

Ne 1

Topology

All topologies 222

Inclusive 60

CC0 π 24CC1 π^+ 24

CCQE 22

CC1 π^0 16NC1 π^0 13

NCE 10

CC1 π^- 10CCcoh1 $\pi^\pm$ 10

CEvNS 7

CC0 π 1p 7NCcoh1 π^0 6NC1 π^+ 5

CC strange 4

CC2 π 4CC1K $^+$ 3NC1 γ 3NC1 π^- 3CC0 π 2p 2

Search title, author, experiment, topology, bibtag...

Sort

Newest first

.bib 222

222 papers

Measurement of coherent elastic neutrino nucleus scattering on germanium by COHERENT

arXiv ↗

INSPIRE ↗

Cite

6

COHERENT

2026

NC

 ν_e ν_μ $\bar{\nu}_\mu$

Ge

CEvNS

Total

Dataset

Measurement of muon (anti-)neutrino charged-current quasielastic-like cross section using off-axis NuMI beam at ICARUS

arXiv ↗

INSPIRE ↗

Cite

1

ICARUS

2026

CC

 ν_μ $\bar{\nu}_\mu$

Ar

CC0 π

Single-diff

First measurement of differential cross sections for muon neutrino charged current interactions on argon with a two-proton final state using the MicroBooNE detector

arXiv ↗

DOI ↗

INSPIRE ↗

Cite

20

MicroBooNE

2026

CC

 ν_μ

Ar

CC0 π 2p

Single-diff

Dataset

Measurement of single charged pion production in charged-current ν_μ -Ar interactions with the MicroBooNE detector

arXiv ↗

DOI ↗

INSPIRE ↗

Cite

5

MicroBooNE

2026

CC

 ν_μ

Ar

CC1 π^- CC1 π^+

Single-diff

Dataset

Measurements of the electron neutrino-argon differential cross section without pions in the final state in MicroBooNE

arXiv ↗

INSPIRE ↗

Cite

1

MicroBooNE

2026

CC

 ν_e

Ar

CC0 π

Single-diff

Dataset

6 papers

CC

Ar

CC0 π

Measurement of muon (anti-)neutrino charged-current quasielastic-like cross section using off-axis NuMI beam at ICARUS

arXiv

INSPIRE

Cite

1

ICARUS

2026

CC

ν_μ

$\bar{\nu}_\mu$

Ar

CC0 π

Single-diff

Measurements of the electron neutrino-argon differential cross section without pions in the final state in MicroBooNE

arXiv

INSPIRE

Cite

1

MicroBooNE

2026

CC

ν_e

Ar

CC0 π

Single-diff

Dataset

Measurement of single- and double-differential cross sections for mesonless charged-current muon neutrino interactions on argon with final-state protons using the MicroBooNE detector

arXiv

DOI

INSPIRE

Cite

19

MicroBooNE

2025

CC

ν_μ

Ar

CC0 π

Double-diff

Dataset

Measurement of charged-current muon neutrino-argon interactions without pions in the final state using the MicroBooNE detector

arXiv

DOI

INSPIRE

Cite

8

MicroBooNE

2025

CC

ν_μ

Ar

CC0 π

Single-diff

Dataset

First Double-Differential Measurement of Kinematic Imbalance in Neutrino Interactions with the MicroBooNE Detector

arXiv

DOI

INSPIRE

Cite

40

MicroBooNE

2023

CC

ν_μ

Ar

CC0 π

Double-diff

Dataset

Differential Cross Section Measurement of Charged Current ν_e Interactions Without Pions in MicroBooNE

arXiv

DOI

INSPIRE

Cite

29

MicroBooNE

2022

CC

ν_e

Ar

CC0 π

Single-diff

Dataset

More about this later

C. JESÚS-VALLS

NUWRO WORKSHOP 2026

14

A QUICK LOOK

Interactions 222

Oscillations 204

Rows

One per result

One per paper

Filter

Experiment

All experiments

Current

CC + NC

Flavor

All flavors

Target

All targets

Topology

All topologies

Measurement

All types

Clear

Columns

Experiment

Year

Key

Citation

Current

Flavour

Target

Topology

Type

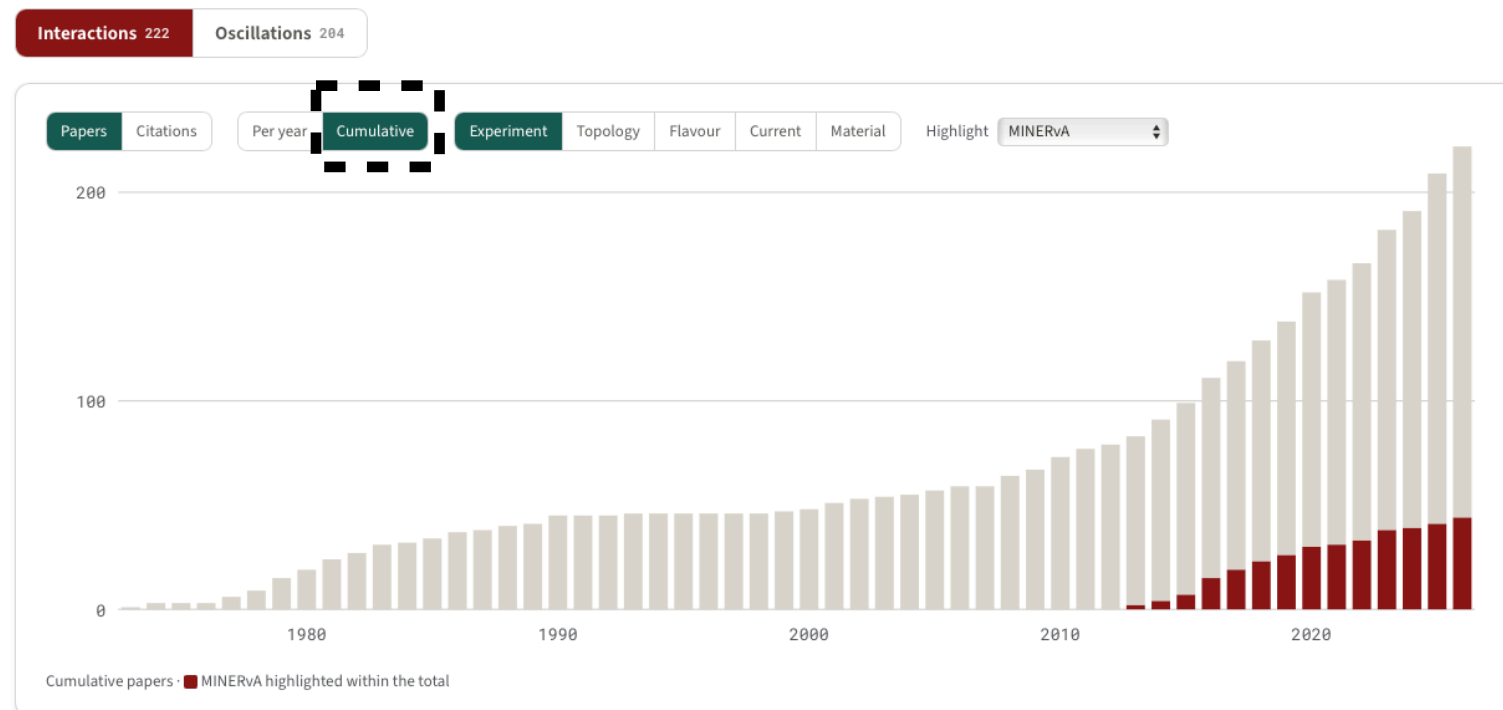
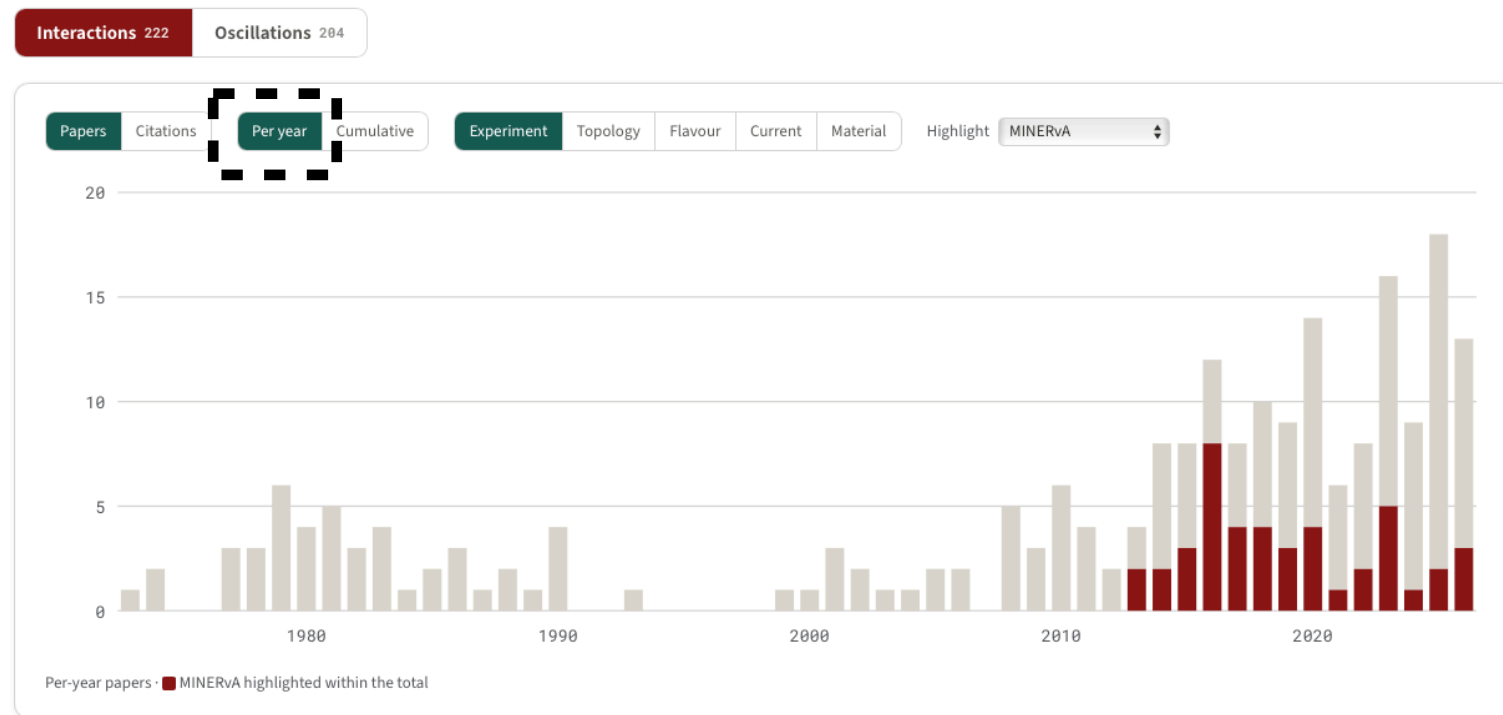
Observable

252 results · 5 columns

Copy LaTeX

.tex

Experiment	Year	Flavour	Target	Topology
COHERENT	2026	$\nu\mu, \bar{\nu}\mu, \nu e$	Ge	CE ν NS
ICARUS	2026	$\nu\mu, \bar{\nu}\mu$	Ar	CC0 π
MicroBooNE	2026	$\nu\mu$	Ar	CC0 π 2p
MicroBooNE	2026	$\nu\mu$	Ar	CC1 π^+
MicroBooNE	2026	$\nu\mu$	Ar	CC1 π^-
MicroBooNE	2026	νe	Ar	CC0 π
MicroBooNE	2026	$\nu\mu$	Ar	CCcoh1 $\pi^\pm$
MINERvA	2026	$\bar{\nu}\mu$	C, CH, Fe, Pb	Inclusive
MINERvA	2026	$\nu\mu$	CH	CC0 π
MINERvA	2026	$\nu\mu$	CH	CC1 π^+
NOvA	2026	$\bar{\nu}\mu$	CH ₂	CC1 π^0
NOvA	2026	$\bar{\nu}\mu$	CH ₂	Inclusive
SND@LHC	2026	$\nu\mu, \bar{\nu}\mu$	W	Inclusive
Super-Kamiokande	2026	$\nu\mu, \bar{\nu}\mu$	Rock	Inclusive
COHERENT	2025	$\nu\mu, \bar{\nu}\mu, \nu e$	Ge	CE ν NS
FASER	2025	$\nu\mu, \bar{\nu}\mu$	W	Inclusive
MicroBooNE	2025	$\nu\mu$	Ar	Inclusive
MicroBooNE	2025	$\nu\mu$	Ar	NC1 π^0
MicroBooNE	2025	$\nu\mu$	Ar	CC0 π
MicroBooNE	2025	$\nu e, \bar{\nu} e$	Ar	CC0 π Np
MicroBooNE	2025	$\nu\mu$	Ar	CC1K $^+$
MicroBooNE	2025	$\nu\mu$	Ar	CC0 π



- Sadly, existing “chaos” in dataset distribution to the community made adding datasets not easy to automatise using agents.
- Having a list of papers is a great starting point. Original question was: Why dont ALL papers have associated public data? So I did some classification.
- **A:** Is the paper in NUISANCE? -> Yes -> Add to nubib (Easy with Agents)
- **B:** Check additional datasets in HepData using INSPIRE-HEP API -> Yes -> Add to nubib (A bit more challenging, HepData has some firewalls to prevent actions from Bots...).
- **C:** Check additional datasets in Zenodo -> Yes -> Add to nubib (Zenodo search engine doesnt work well!...)
- **D:** Deal with ALL missing papers, one by one.
 - Large collaborations (e.g. T2K, MINERvA, had some extra results in their webpages, which I could rescue by hand). -> Add to nubib
 - Read all missing papers, many tens (with the help of Agents). Identified papers that had the result in the abstract (e.g. inclusive cross section reported there, upper limit reported there...). -> Add to nubib
 - Made a list of papers with data I couldn't find anywhere, but that wasn't trivially in the abstract. Decide I wanted to digitise ALL of them...

In addition I created some standard formats (some “recipes”) to store datasets and visualise them on the web
Lots of customisation needed to fit all the particularities in each experiment.

https://github.com/cesarjesusvalls/NuIntBib/blob/main/scripts/build_datasets.py

In total nubib has now datasets for 201 papers!

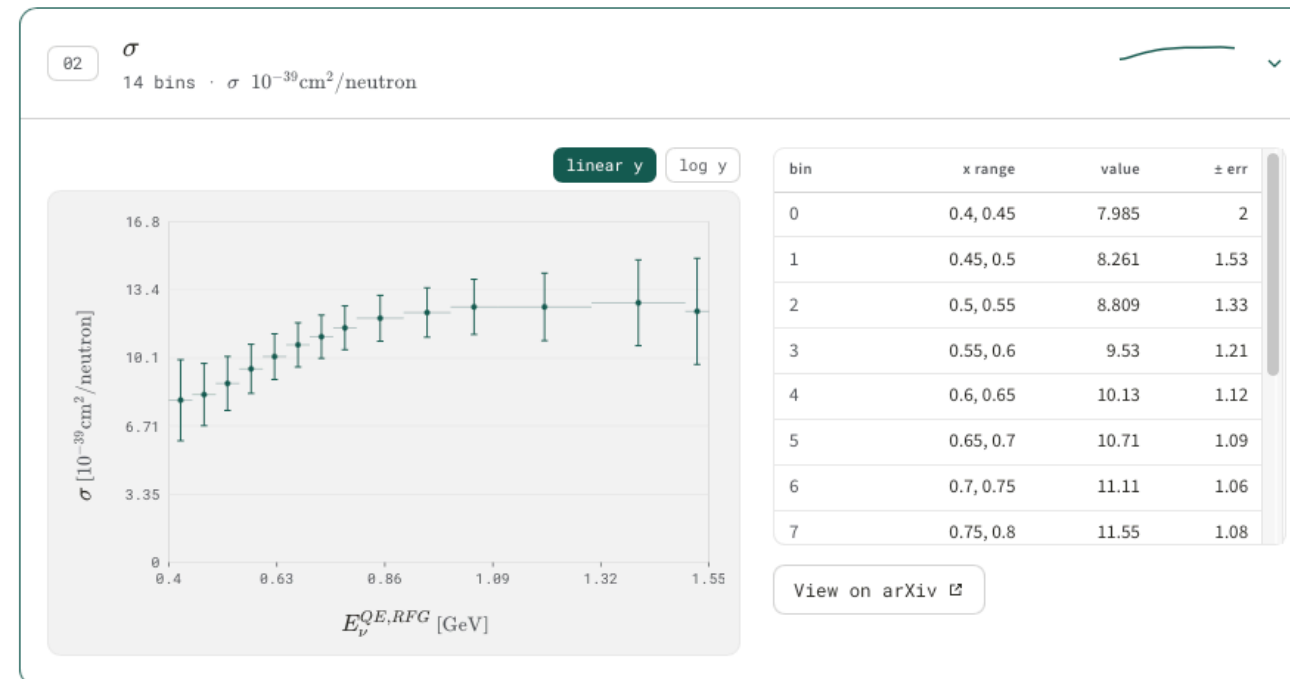
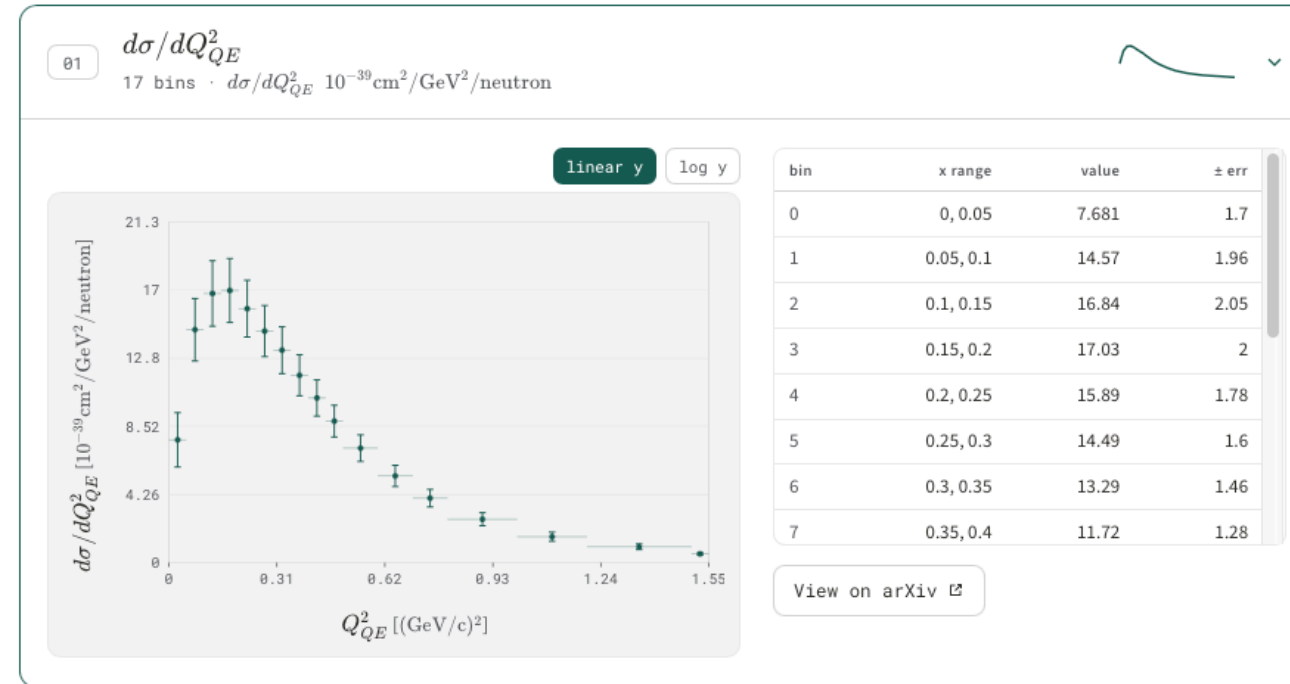
Labels

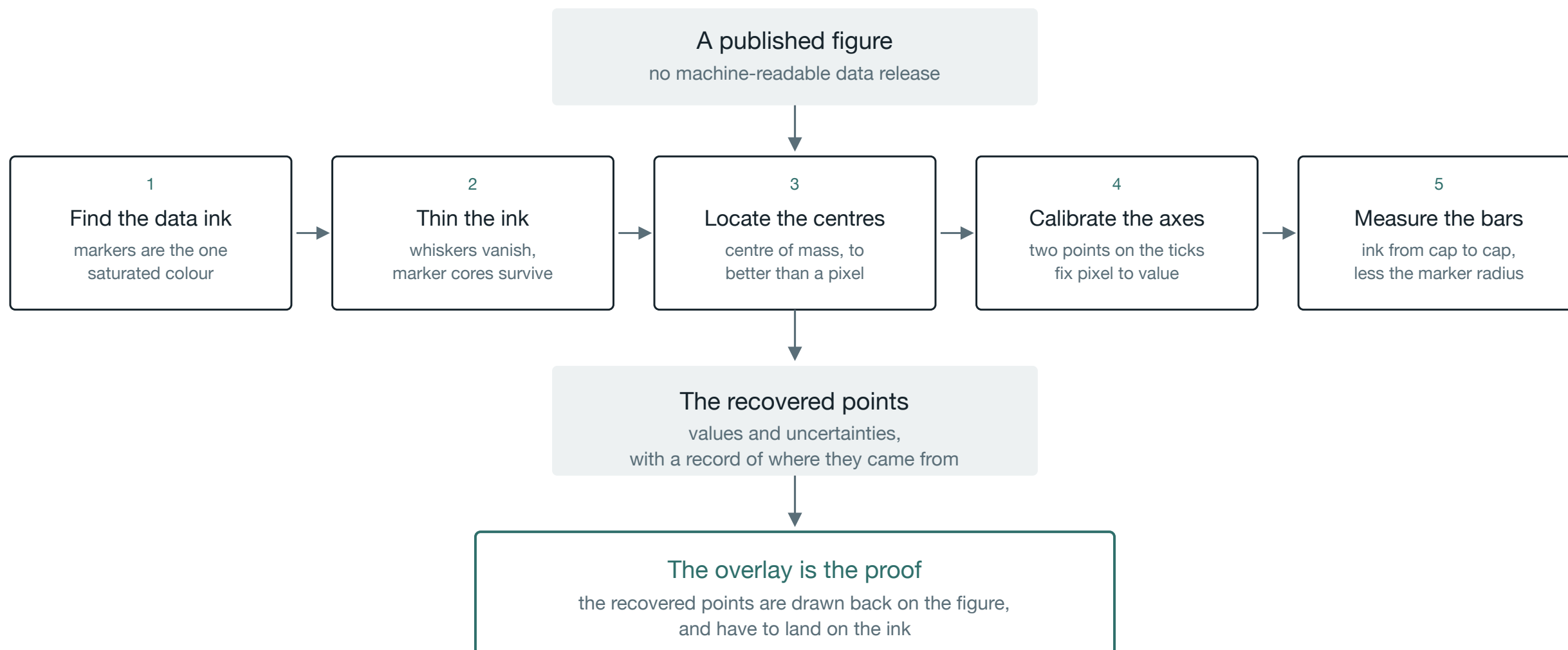
CC ν_μ C CCQE Double-diff

Data release

↓ Download files (ZIP)

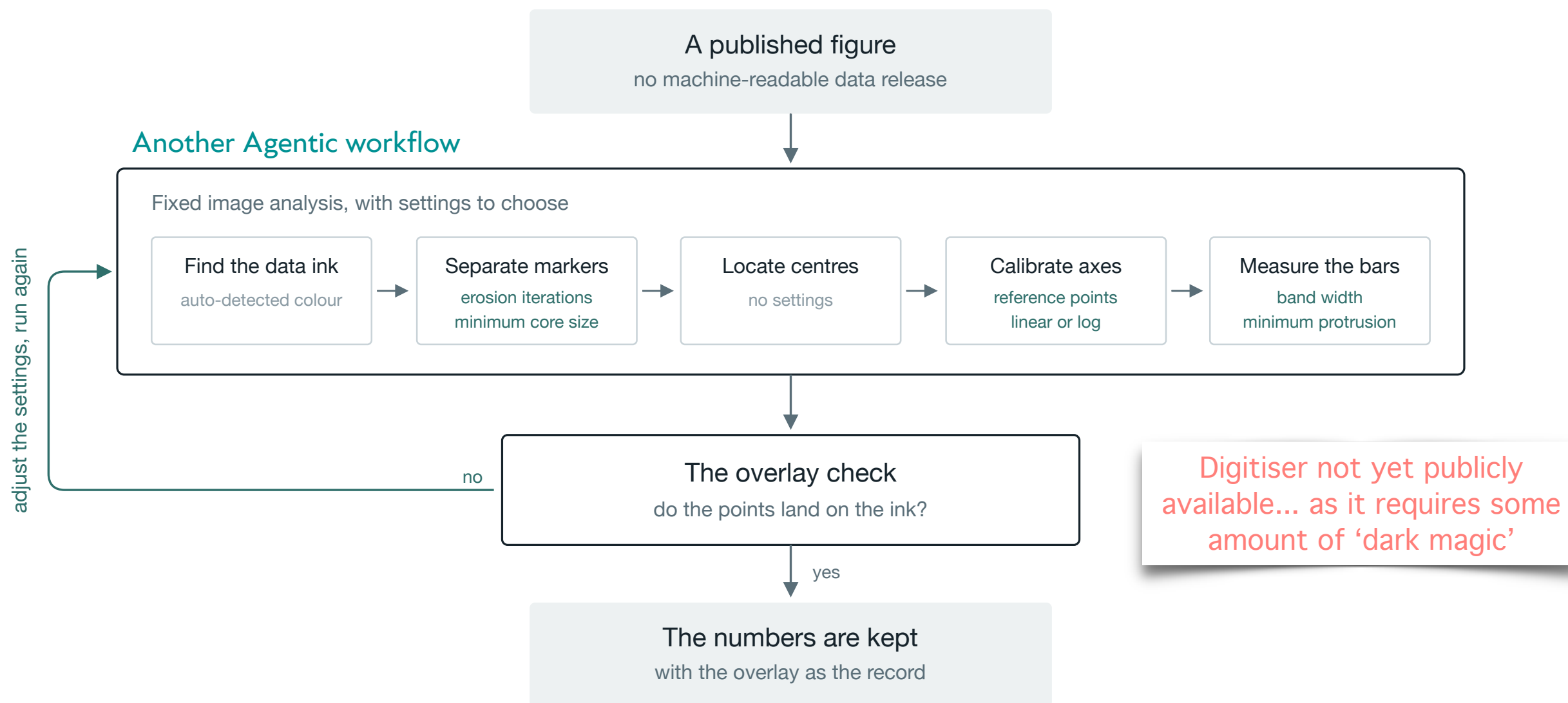
Release from [arXiv](#).





Classical image analysis throughout: no model, no machine learning, nothing eyeballed.

Module	Key functions	Does
<code>detect.py</code>	<code>detect_markers(mask, erode_iter=2, min_core_area=4)</code>	data-ink isolation + marker/whisker separation → one core per point
<code>calibrate.py</code>	<code>calibrate_axis(...)</code>	pixel↔data map from reference points (linear/log)
<code>autocalibrate.py</code>	<code>detect_spines</code> , <code>detect_ticks</code> , <code>detect_labels</code> , <code>detect_multiplier</code> , <code>calibrate_axis</code>	auto -finds the axes, ticks, and tick labels — so calibration can happen without you hand-clicking references
<code>errorbars.py</code>	<code>measure(mask, marker, calib, band=3, min_protrusion=3.0)</code>	error-bar cap-to-cap ink run → x/y uncertainty
<code>axes.py</code>	axis/scale handling	log vs linear axis transforms



DIGITIZED FROM THE PAPER FIGURE

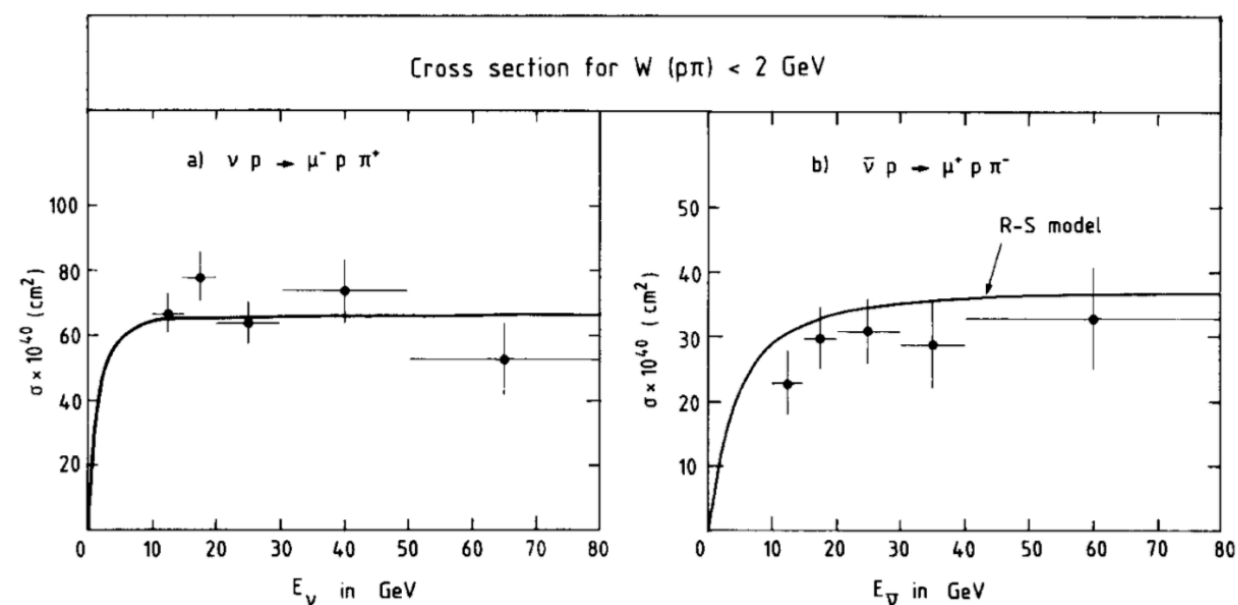


Fig. 3. Cross sections for production of $(p\pi)$ states with effective mass $W < 2$ GeV as functions of the incident (anti)neutrino energy, (a) in reaction (1) and (b) in reaction (2).

Paper figure

incident (anti)neutrino energy, (a) in reaction (1) and (b) in reaction (2).

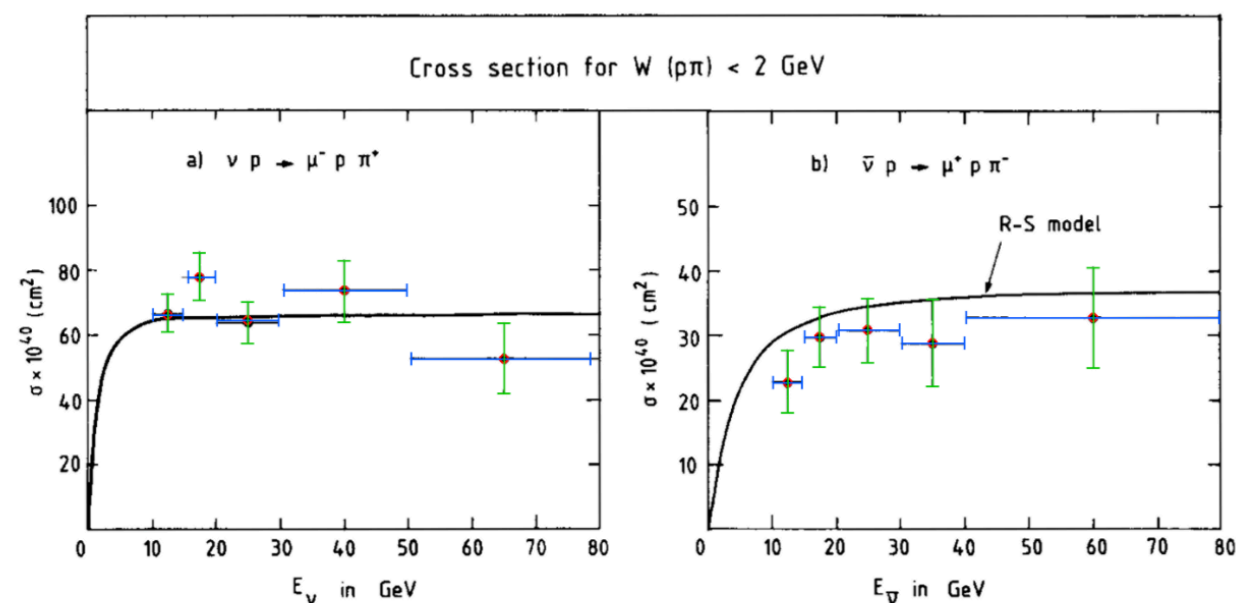
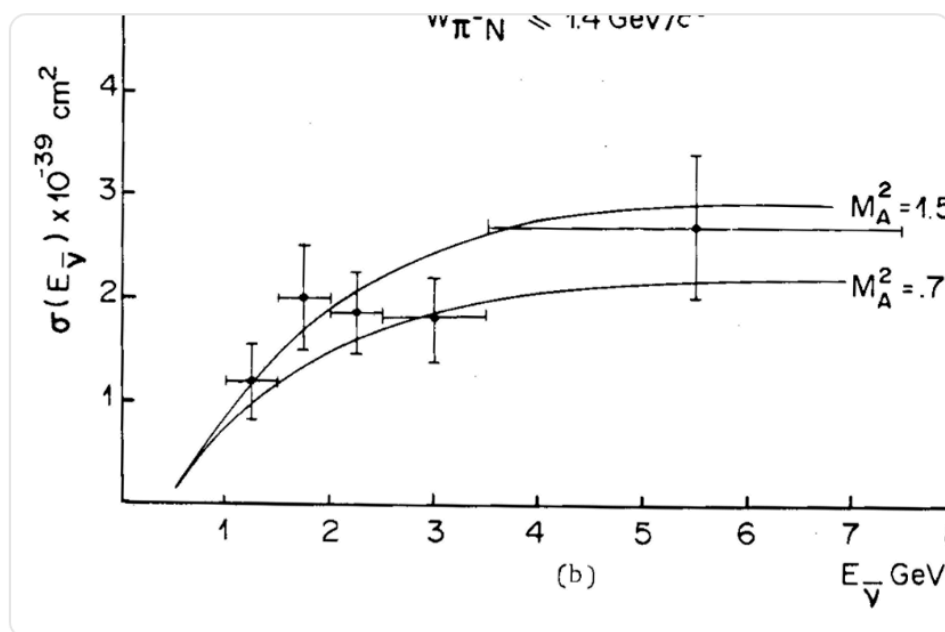


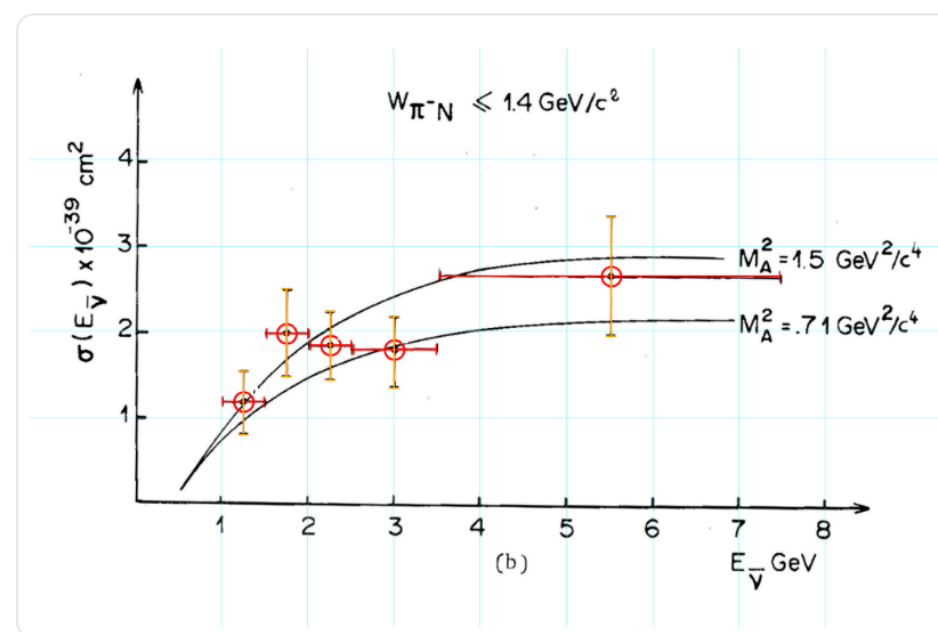
Fig. 3. Cross sections for production of $(p\pi)$ states with effective mass $W < 2$ GeV as functions of the incident (anti)neutrino energy, (a) in reaction (1) and (b) in reaction (2).

Our extraction overlaid

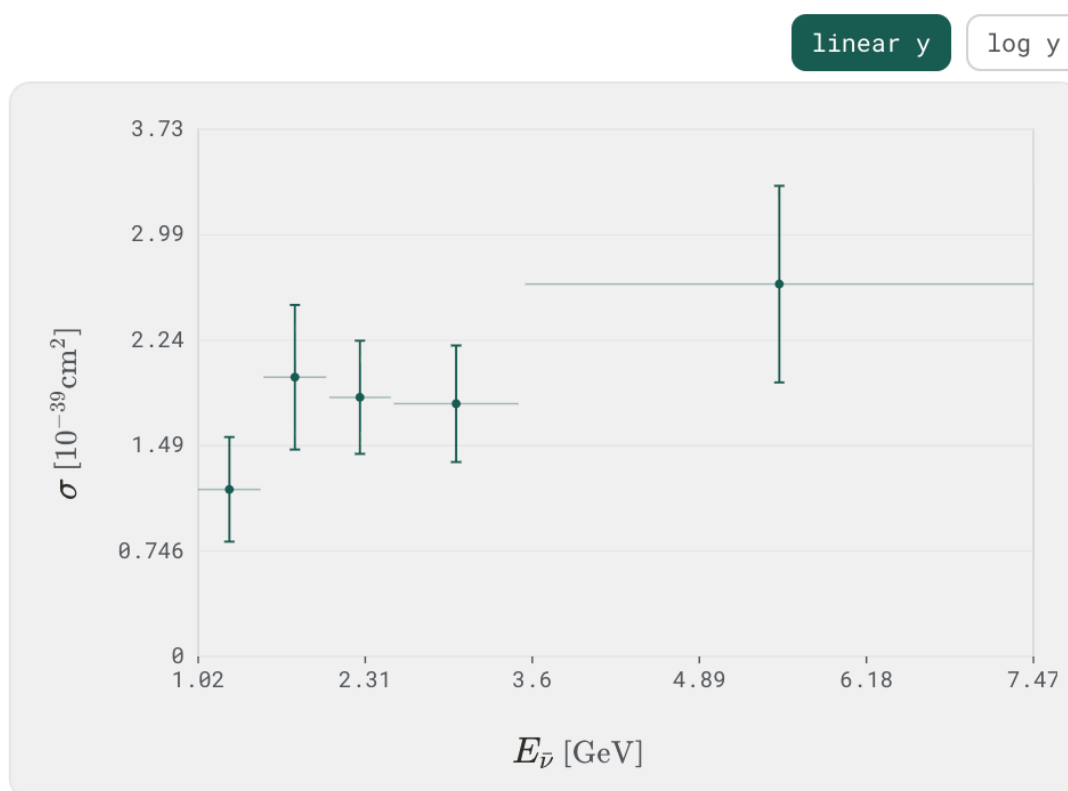
DIGITIZED FROM THE PAPER FIGURE



Paper figure



Our extraction overlaid



bin	x range	value	$\pm \text{err}$
0	1.023, 1.504	1.183	0.37
1	1.529, 2.01	1.977	0.512
2	2.036, 2.508	1.835	0.4
3	2.534, 3.496	1.789	0.413
4	3.548, 7.473	2.636	0.696

[View on published](#)

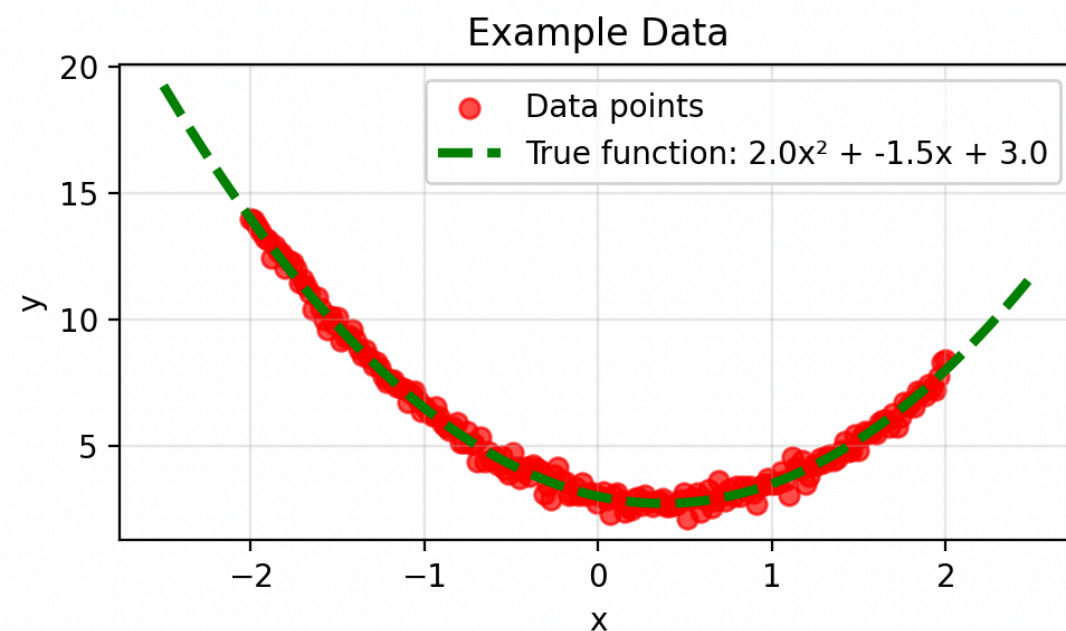
So far...

- v0 -----> nu interaction papers*
- v1 -----> added neutrino oscillation papers*
- v2 -----> added neutrino interaction datasets*
- v3 -----> nu interaction theory papers? [I need feedback and advice!]*

e.g. what would be a relevant set of tags?

Changing topics:
ADoNIS

Let's consider a simple example: A 2nd degree polynomial fit.



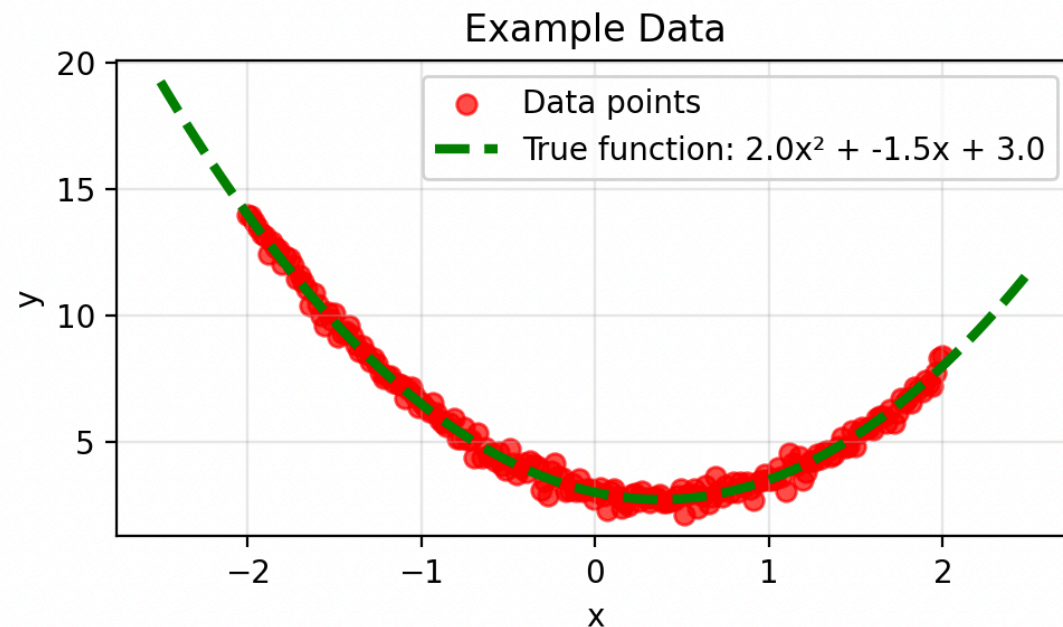
Consider a set of observations generated by $f(x) = ax^2 + bx + c$.
Find the “**best**” (a, b, c) to explain this data.

First we need to define what “**best**” means → define the **loss** $\mathcal{L}$

We can try to find a solution in many ways:

- A) Stochastic Search Methods.
- B) Numerically → $\partial\mathcal{L}/\partial a \approx [\mathcal{L}(a + h, b, c) - \mathcal{L}(a, b, c)]/h$
- C) Analytically → Calculating explicitly $\partial\mathcal{L}/\partial a$

Let's consider a simple example: A 2nd degree polynomial fit.



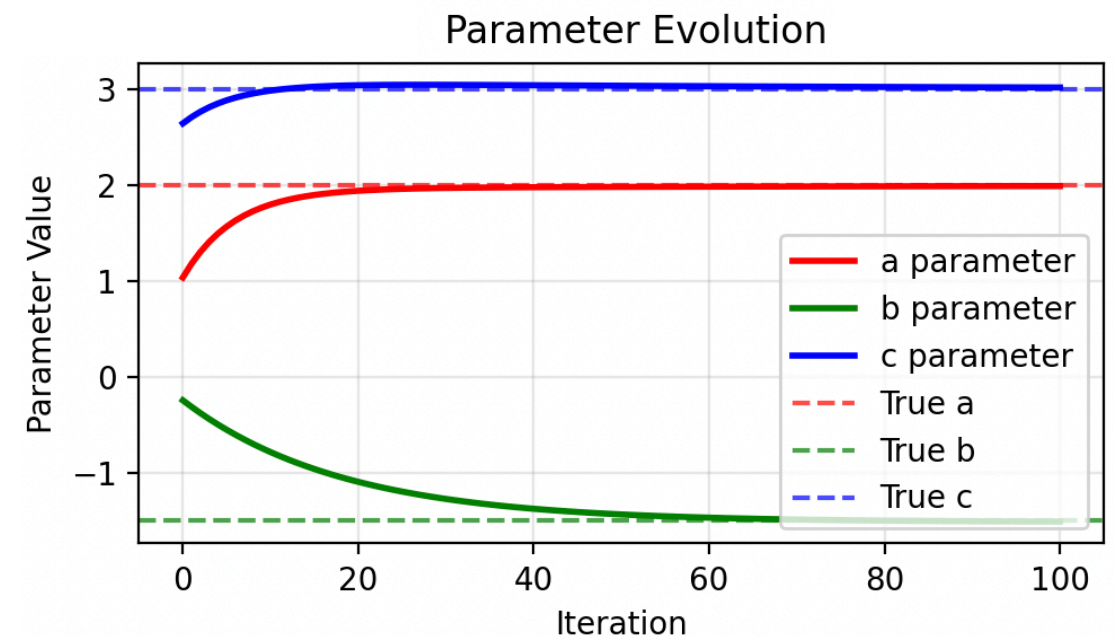
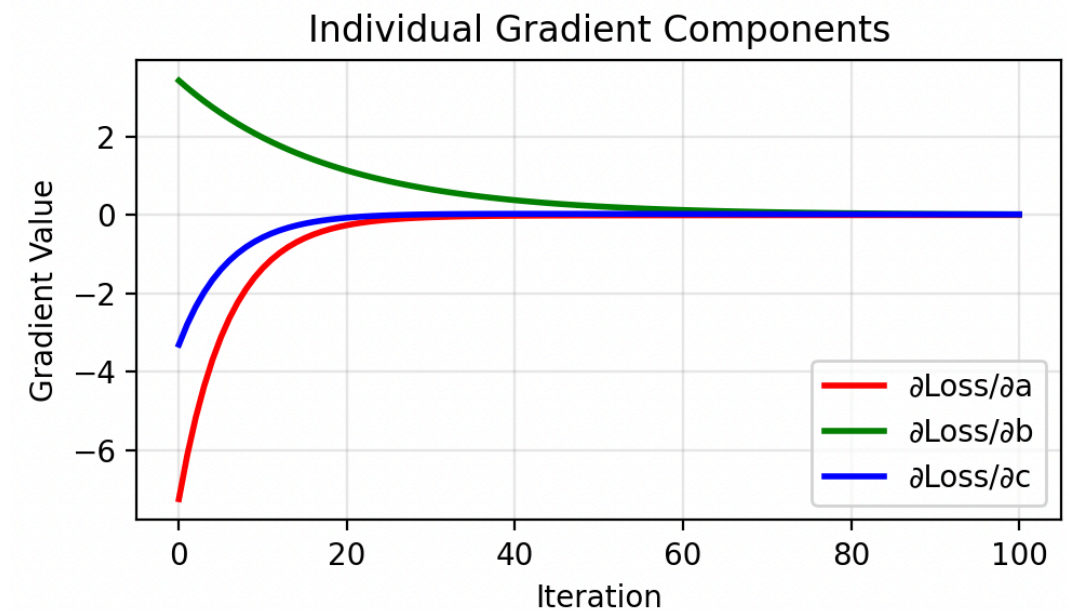
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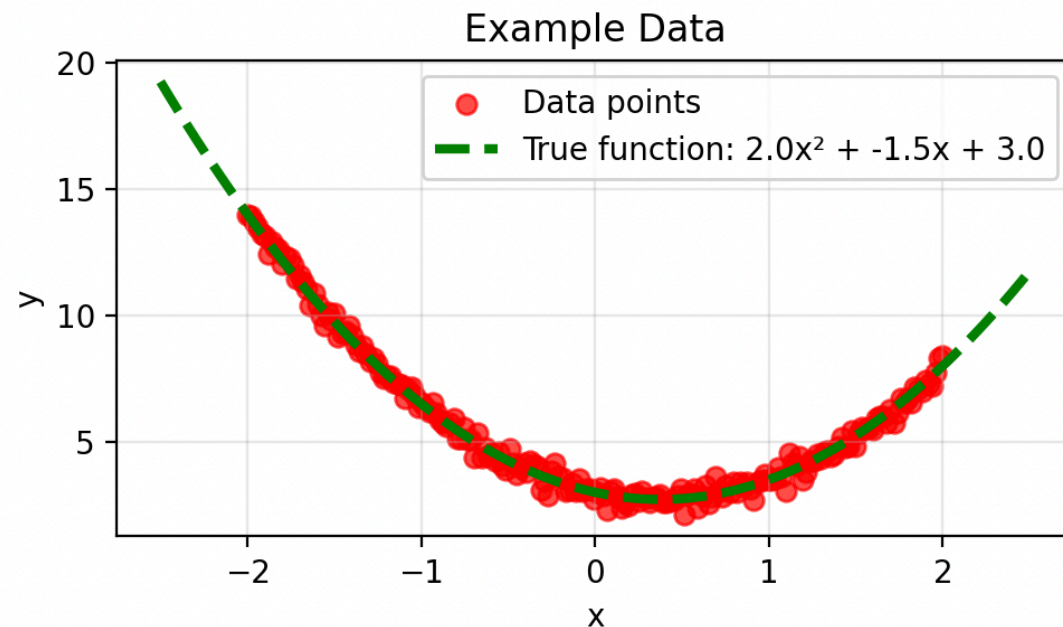
We can try to find a solution in many ways:

- A) Stochastic Search Methods.
- B) Numerically → $\partial\mathcal{L}/\partial a \approx [\mathcal{L}(a + h, b, c) - \mathcal{L}(a, b, c)]/h$
- C) Analytically → Calculating explicitly $\partial\mathcal{L}/\partial a$

Look for a solution iteratively $a_{i+1} = a_i + \alpha \cdot \partial\mathcal{L}/\partial a$ (where α is a scaling variable —learning rate—).



Let's consider a simple example: A 2nd degree polynomial fit.



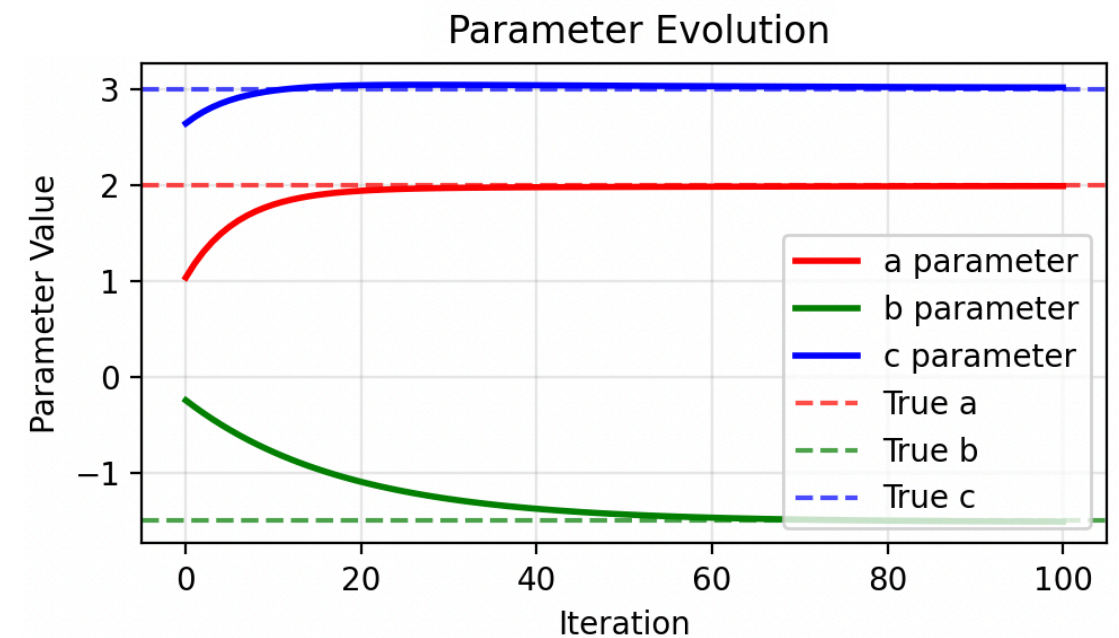
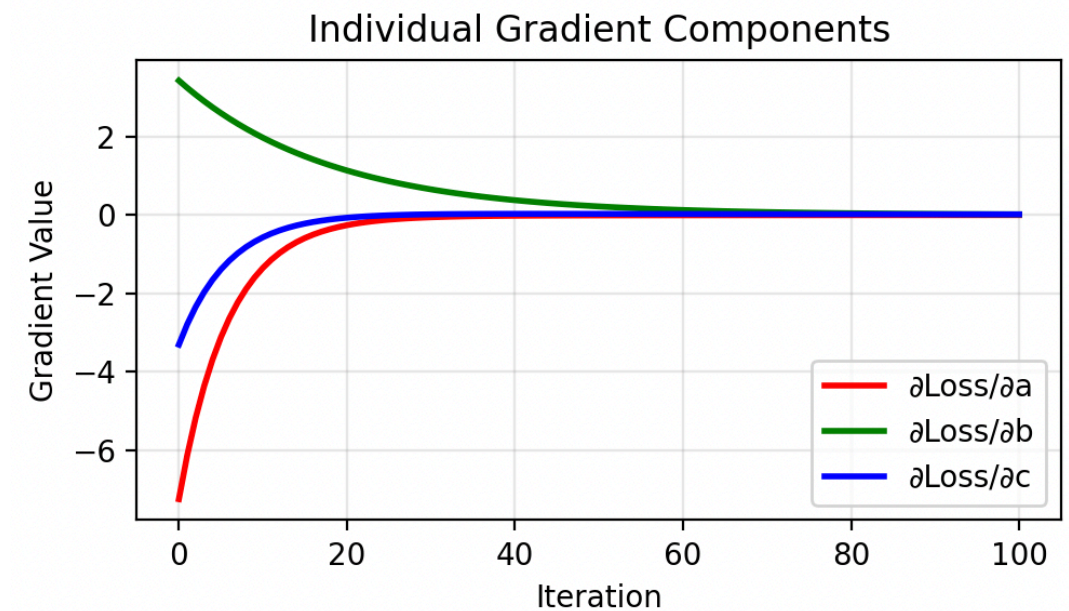
Consider a set of observations generated by $f(x) = ax^2 + bx + c$. Find the “**best**” (a, b, c) to explain this data.

First we need to define what “**best**” means → define the **loss** $\mathcal{L}$

We can try to find a solution in many ways:

- A) Stochastic Search Methods.
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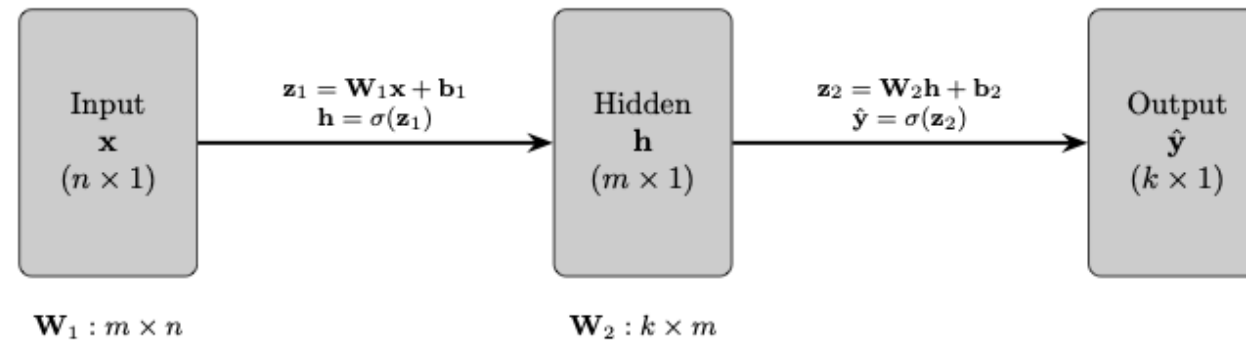
Look for a solution iteratively $a_{i+1} = a_i + \alpha \cdot \partial\mathcal{L}/\partial a$ (where α is a scaling variable —learning rate—).



In a differentiable system, given $f_{\vec{\theta}}(\vec{y}) \rightarrow \vec{x}$, we can calculate $\partial\mathcal{L}/\partial\vec{y}$, and use it to perform gradient-based optimization, automatically providing $g_{\vec{\theta}}(\vec{x}) \rightarrow \vec{y}$

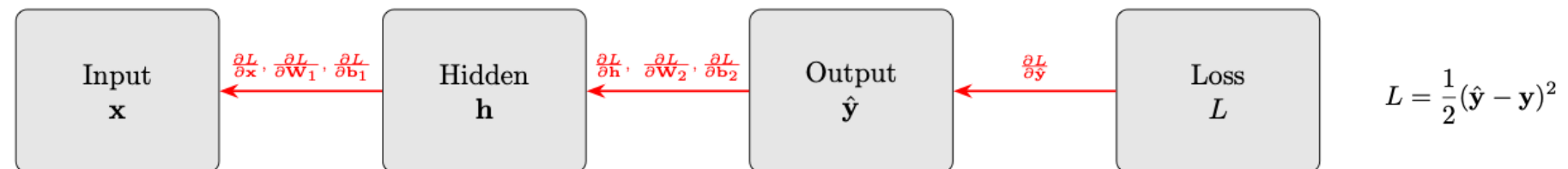
But physics simulators are much more complex than a second degree polynomial, and nobody wants to do all the derivatives by hand!

Automatic gradient calculation is the backbone of neural networks. A simple NN as an example:



To update parameters, we need: $\frac{\partial L}{\partial \mathbf{W}_1}$, $\frac{\partial L}{\partial \mathbf{b}_1}$, $\frac{\partial L}{\partial \mathbf{W}_2}$, $\frac{\partial L}{\partial \mathbf{b}_2}$

Backpropagation



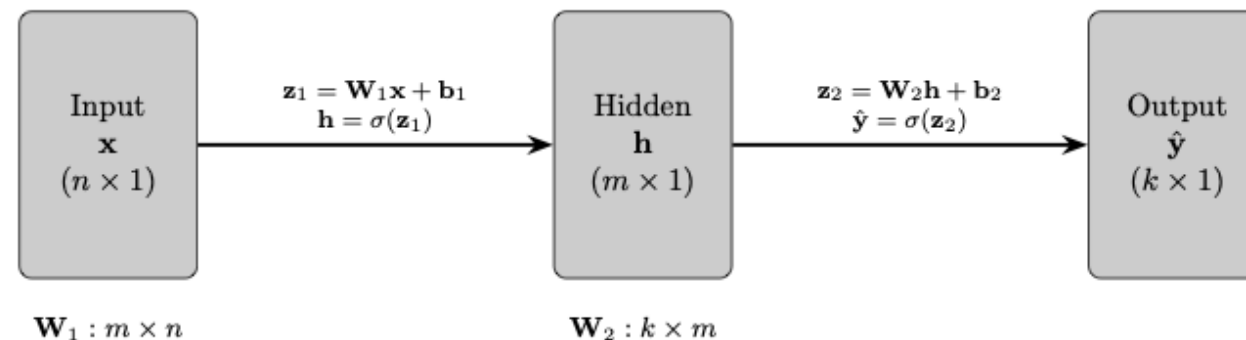
Modern neural networks can be very sophisticated and have an arbitrary number of parameters.

Existing libraries are able to calculate gradients for all their parameters via the chain rule.

GPU-acceleration makes this calculations exceptionally fast.

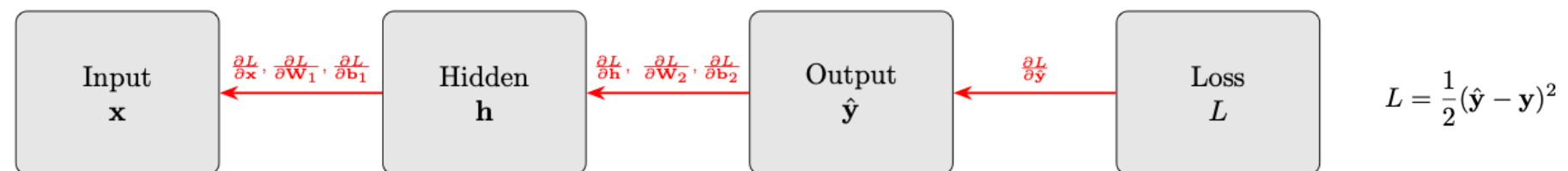
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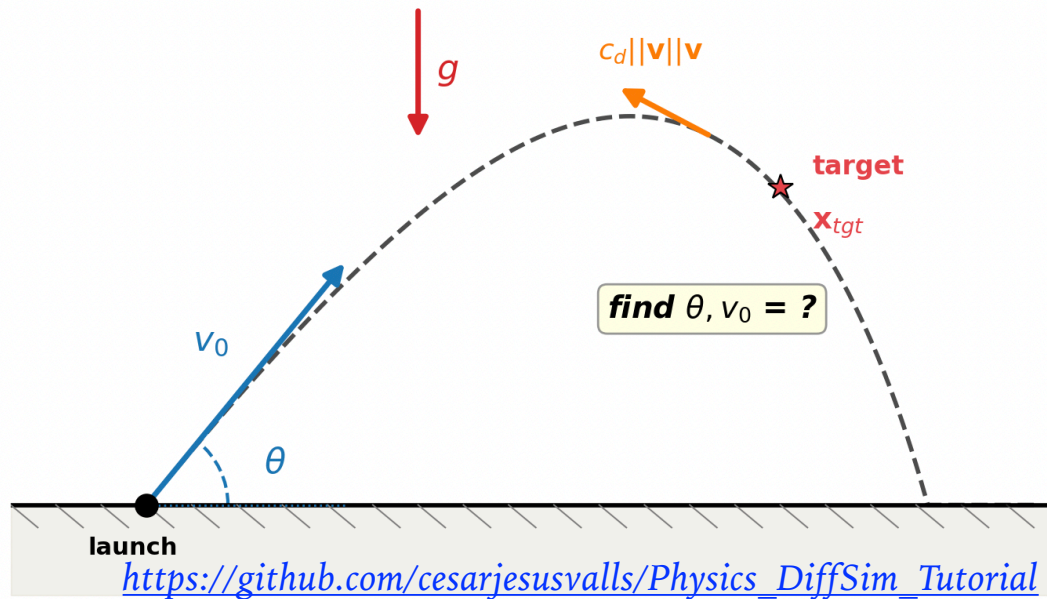
GPU-acceleration makes this calculations exceptionally fast.

Does this mean that we need to use neural networks? No!

We can use differentiable libraries to calculate gradients for any set of composable differentiable functions

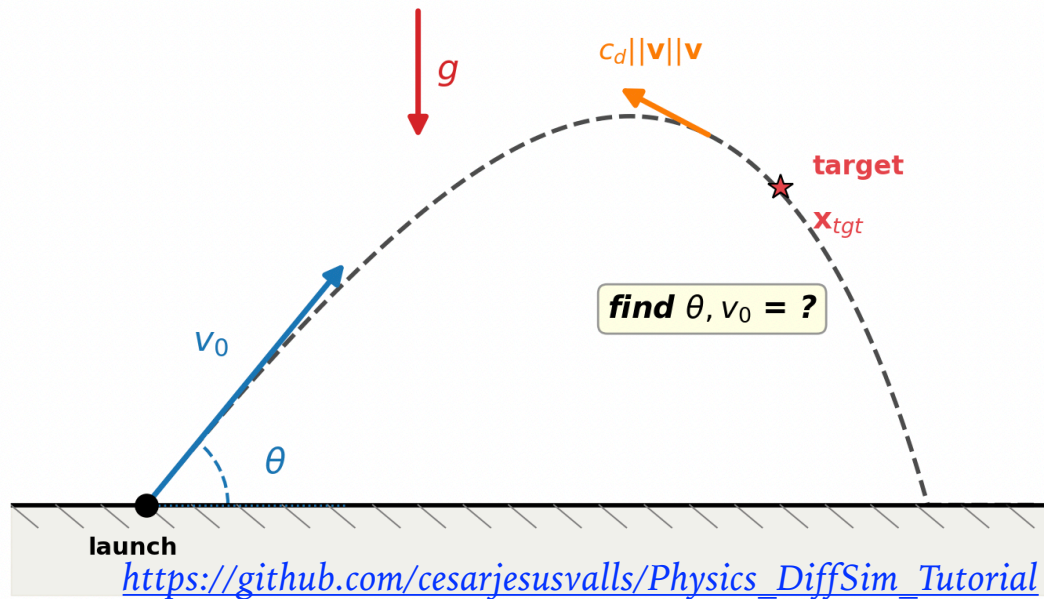
What is a *differentiable physics simulation*?

Inverse problem: optimise launch parameters to hit target



What is a *differentiable physics simulation*?

Inverse problem: optimise launch parameters to hit target



FORWARD

Let's first calculate trajectory taking many small steps forward

$$\mathbf{a}_t = -g \hat{y} - c_d \|\mathbf{v}_t\| \mathbf{v}_t$$

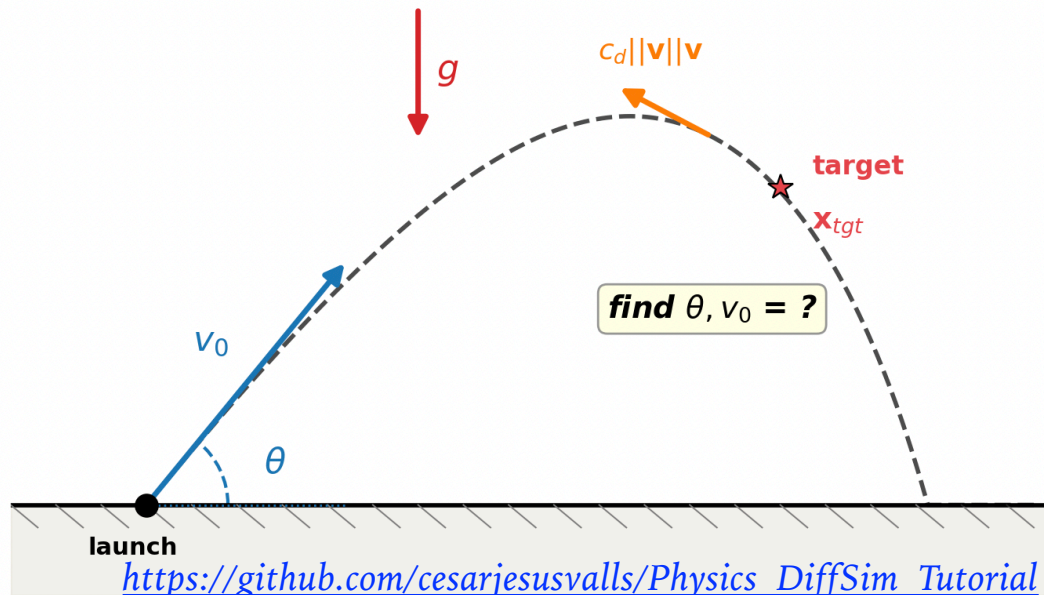
$$\mathbf{v}_{t+1} = \mathbf{v}_t + \mathbf{a}_t \Delta t$$

$$\mathbf{x}_{t+1} = \mathbf{x}_t + \mathbf{v}_{t+1} \Delta t$$

Then let's calculate how far is the closest point in the trajectory to the target, and define the loss $\mathcal{L}$

What is a *differentiable physics simulation*?

Inverse problem: optimise launch parameters to hit target



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Let's first calculate trajectory taking many small steps forward

$$\mathbf{a}_t = -g \hat{y} - c_d ||\mathbf{v}_t|| \mathbf{v}_t$$

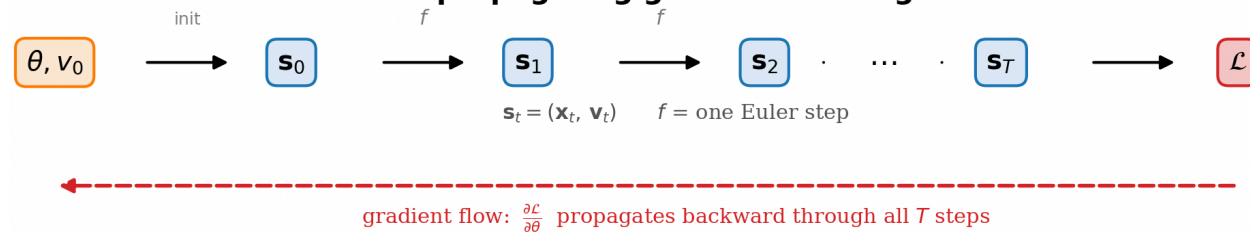
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BACKWARD

Chain rule: backpropagating gradients through the simulation



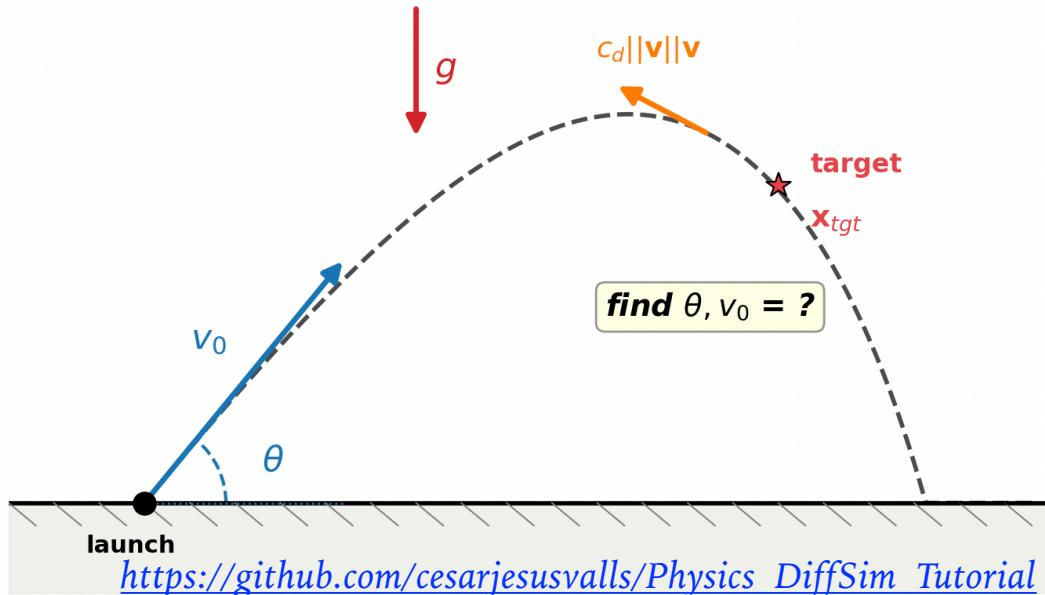
Chain rule (reverse mode AD) :

$$\frac{\partial \mathcal{L}}{\partial \theta} = \frac{\partial \mathcal{L}}{\partial \mathbf{s}_T} \cdot \frac{\partial \mathbf{s}_T}{\partial \mathbf{s}_{T-1}} \cdot \dots \cdot \frac{\partial \mathbf{s}_1}{\partial \mathbf{s}_0} \cdot \frac{\partial \mathbf{s}_0}{\partial \theta}$$

$$= \frac{\partial \mathcal{L}}{\partial \mathbf{s}_T} \cdot \prod_{t=1}^T \frac{\partial f}{\partial \mathbf{s}} |_{\mathbf{s}_{t-1}} \cdot \frac{\partial \mathbf{s}_0}{\partial \theta}$$

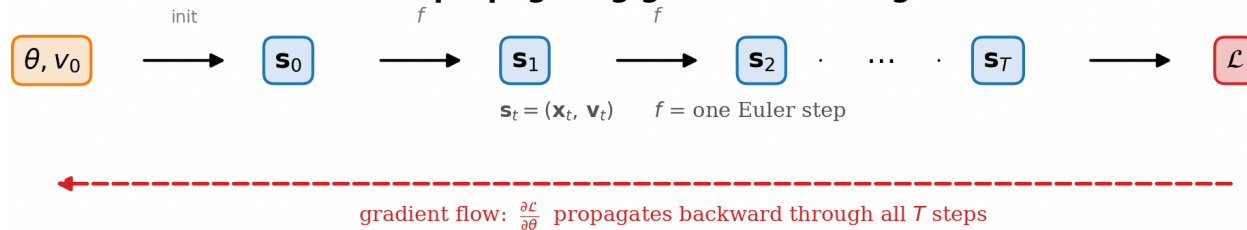
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BACKWARD

Chain rule: backpropagating gradients through the simulation



Chain rule (reverse mode AD):

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Take home message: We define the forward operation, we tune parameters leveraging AD!

FORWARD

Let's first calculate trajectory taking many small steps forward

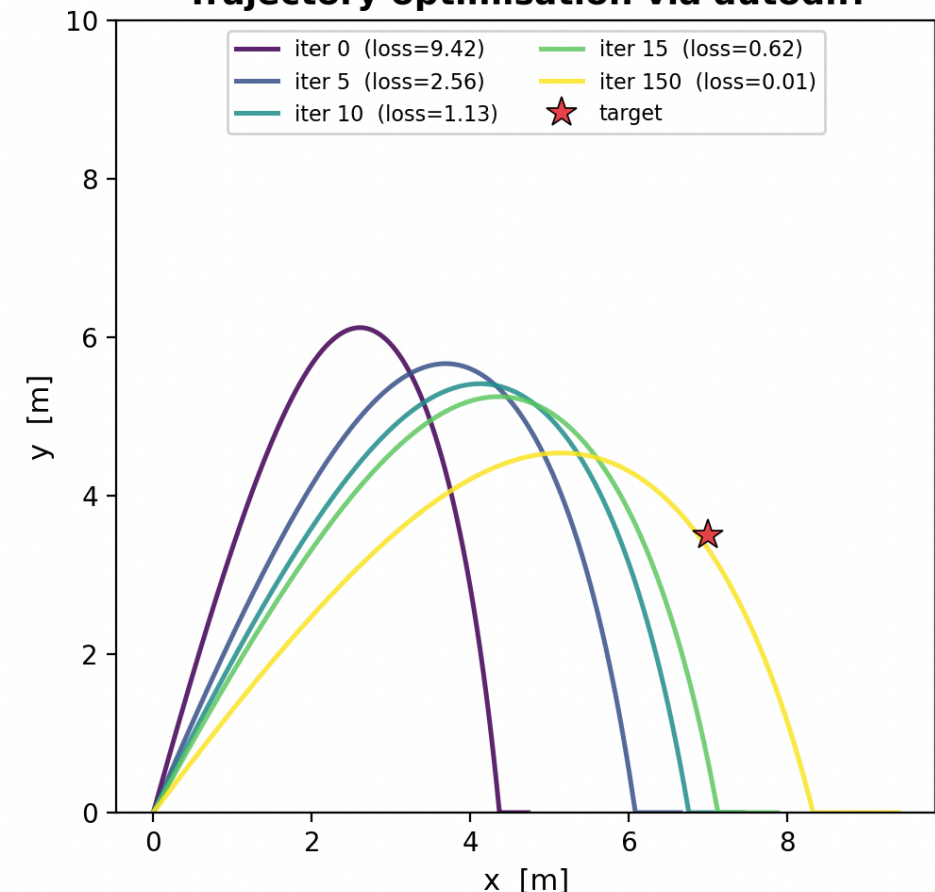
$$\mathbf{a}_t = -g \hat{y} - c_d ||\mathbf{v}_t|| \mathbf{v}_t$$

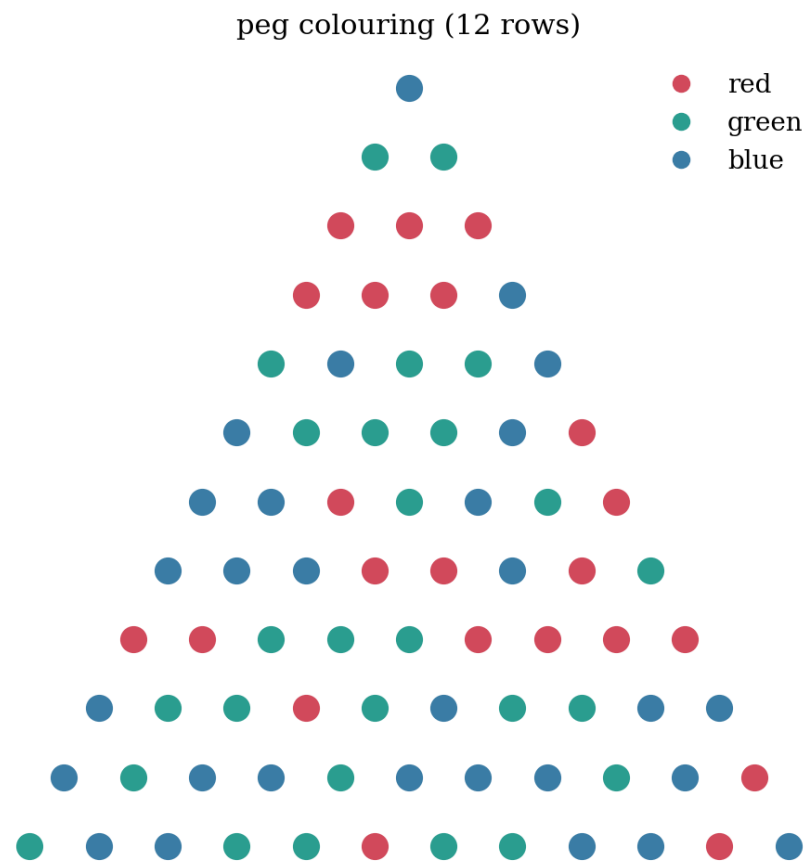
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Then let's calculate how far is the closest point in the trajectory to the target, and define the loss $\mathcal{L}$

Trajectory optimisation via autodiff



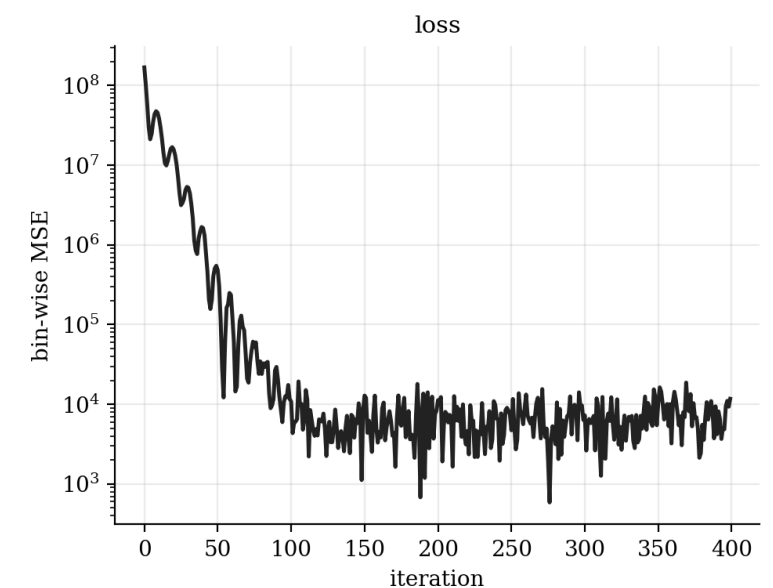
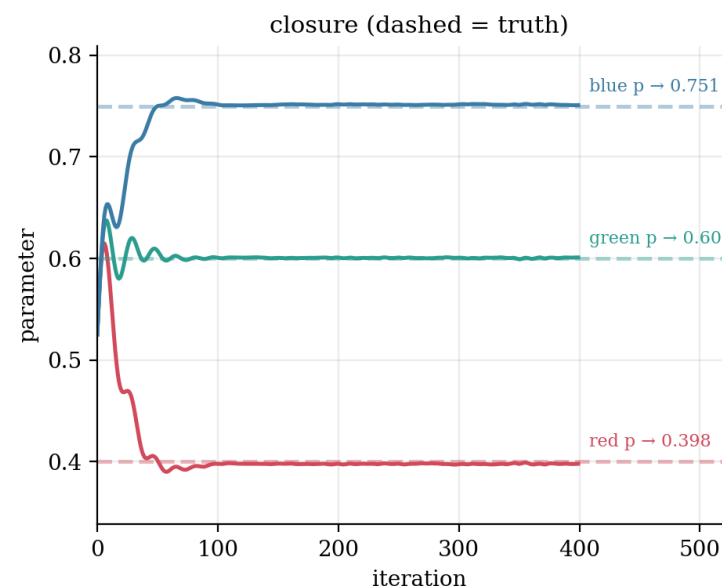
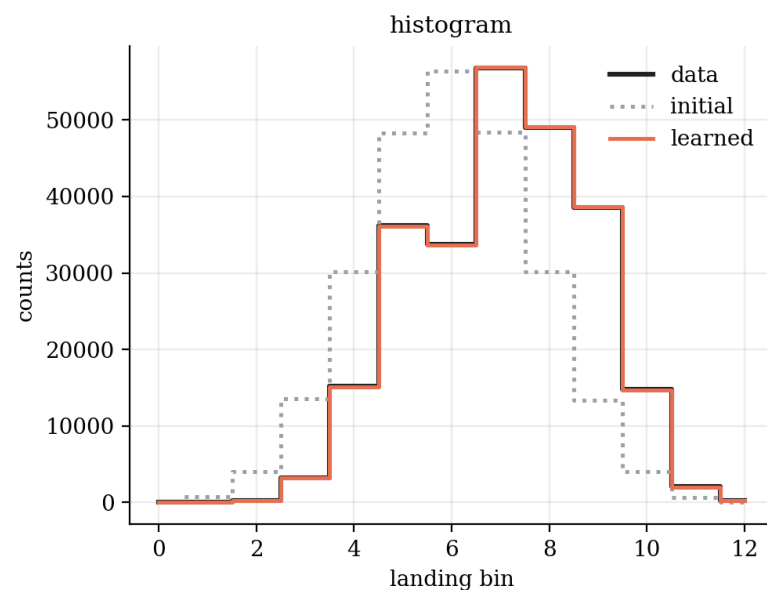


What about systems where stochasticity is at the core?

No problem! One can preserve gradient information throughout!

<https://github.com/cesarjesusvalls/differentiable-sampling>

Galton board: recovering per-colour right-turn probabilities



Can we develop a differentiable neutrino event generator?

ADoNIS: A Differentiable generatOr of Neutrino Interaction Samples

César Jesús-Valls^{1, *}

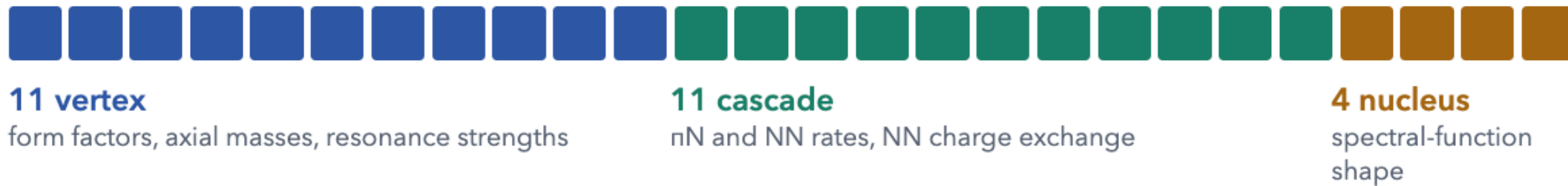
¹*European Organization for Nuclear Research (CERN), 1211 Geneva 23, Switzerland*

Neutrino interaction generators are central to precision oscillation analyses, but conventional implementations do not directly provide derivatives of their predictions with respect to physics parameters, leaving analyses to obtain this dependence through reweighting, finite differences or external response models. We present ADONIS, a fully differentiable neutrino interaction event generator that makes event reweighting and its gradients directly available, propagating exact derivatives through the nuclear ground state, hard-scattering amplitudes and stochastic intranuclear cascade. Following the physics choices of ACHILLES, we demonstrate agreement with its predictions across neutrino, electron and hadron probes, showing that differentiability is achieved without loss of physical fidelity. We then show how the resulting Jacobian exposes which measurements and regions of phase space constrain each parameter and how different probes break degeneracies, providing a quantitative tool for fit design, model tuning, measurement design and experiment design. The same differentiable predictions can be used directly in frequentist and Bayesian inference and carried through detector response to reconstructed observables for unfolding, while also yielding computational gains. In sum, ADONIS provides a practical recipe for constructing differentiable neutrino event generators, applicable beyond the particular physics models implemented here, and this paper demonstrates their use across a wide range of tasks relevant to both theory and experiment.

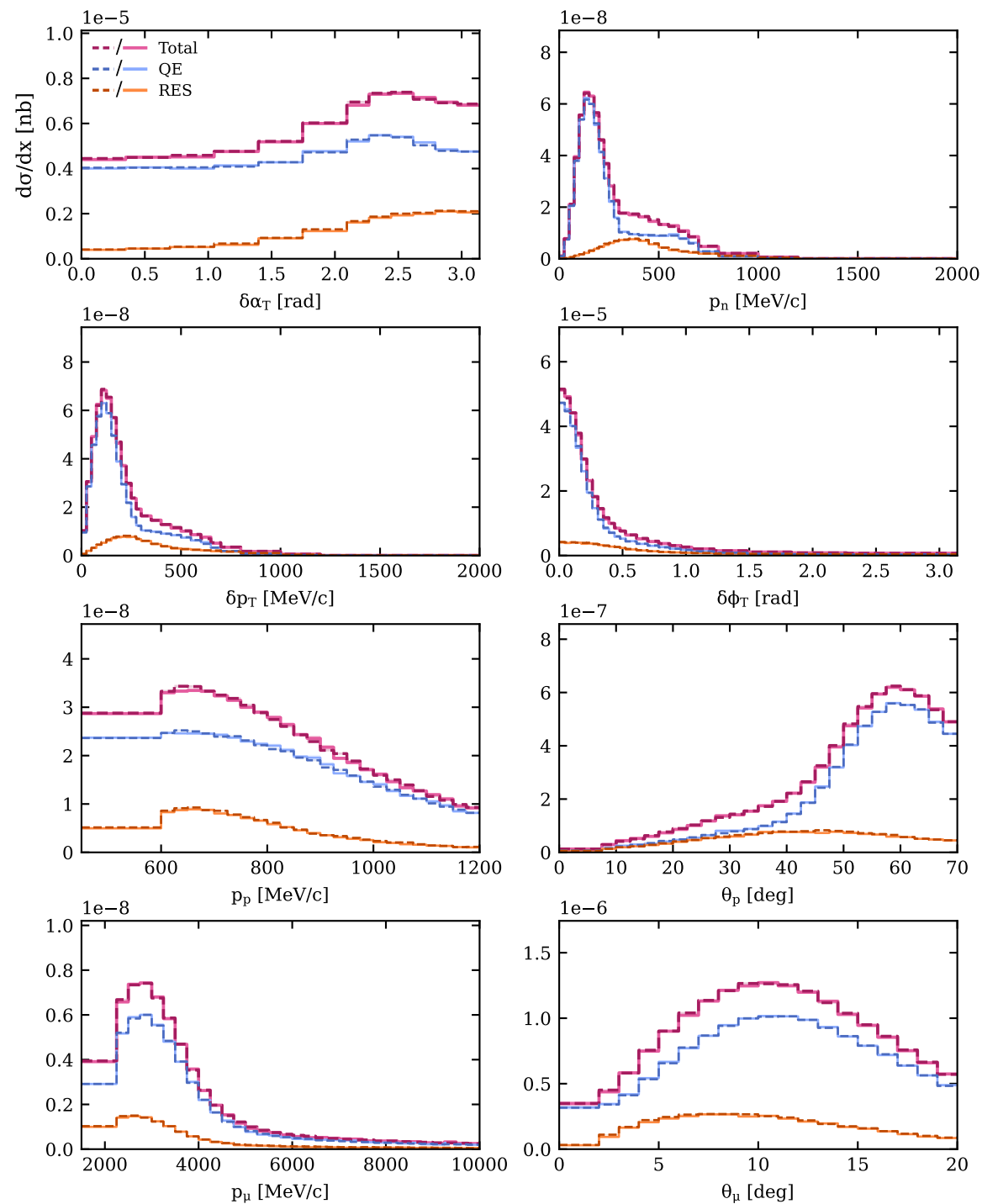
<https://arxiv.org/pdf/2608.25107>

One can preserve gradient information throughout!

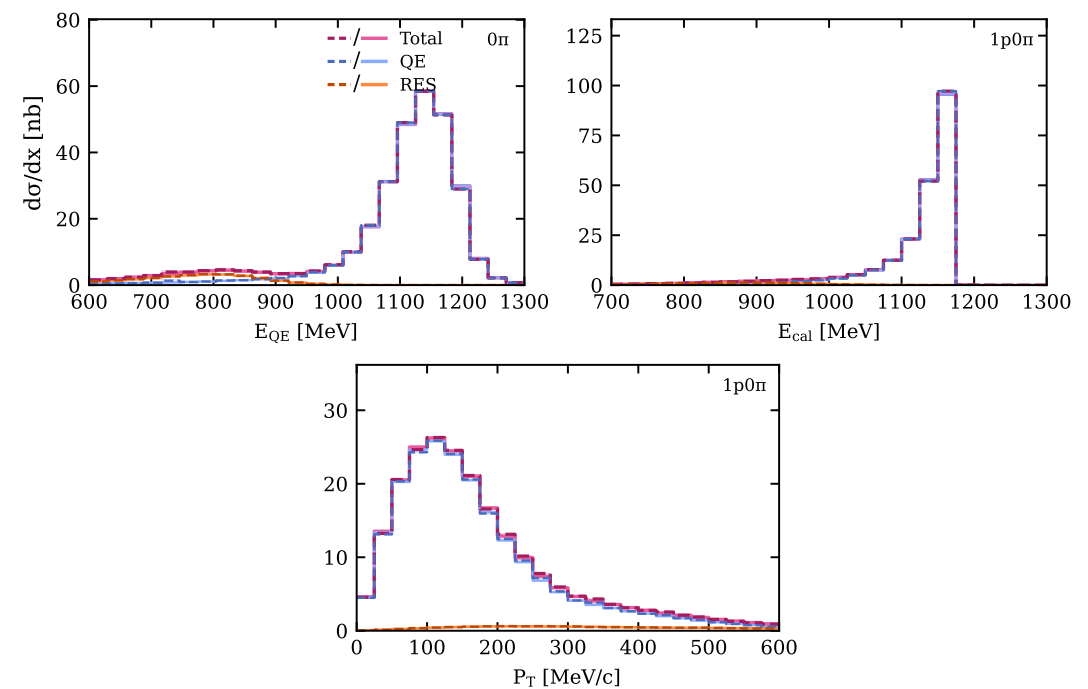
$$w_i(\boldsymbol{\theta}) = w_i^0 r_i^{\text{vertex}}(\boldsymbol{\theta}) r_i^{\text{FSI}}(\boldsymbol{\theta}) r_i^{\text{SF}}(\boldsymbol{\theta})$$



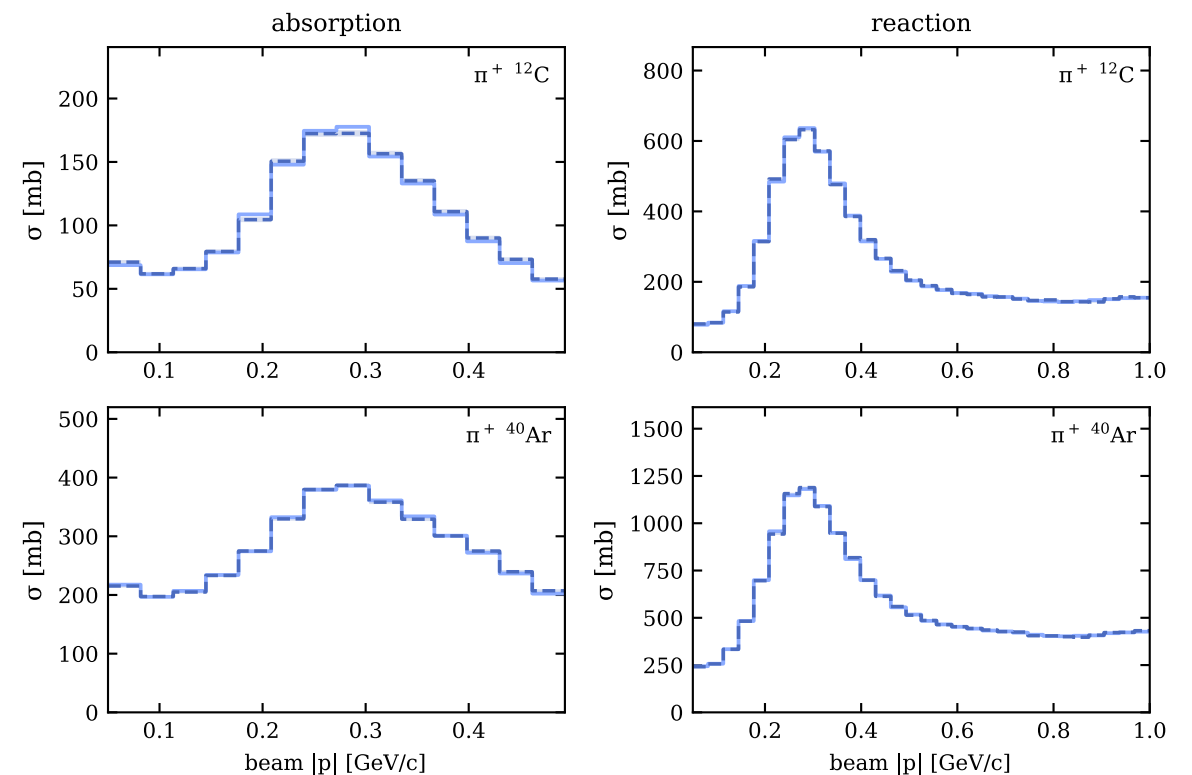
Example neutrino distribution for MINERvA



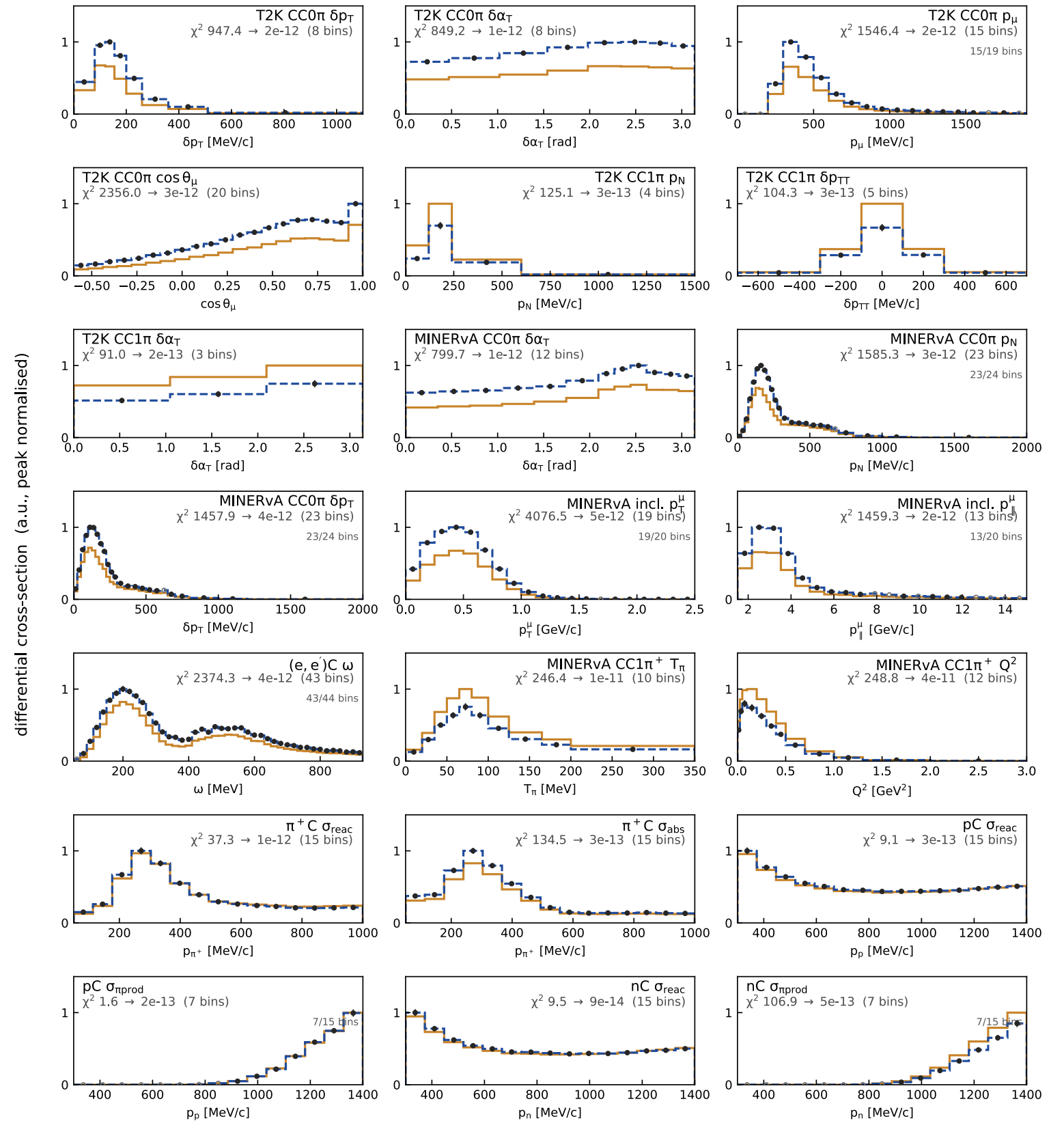
Example electron beam observables



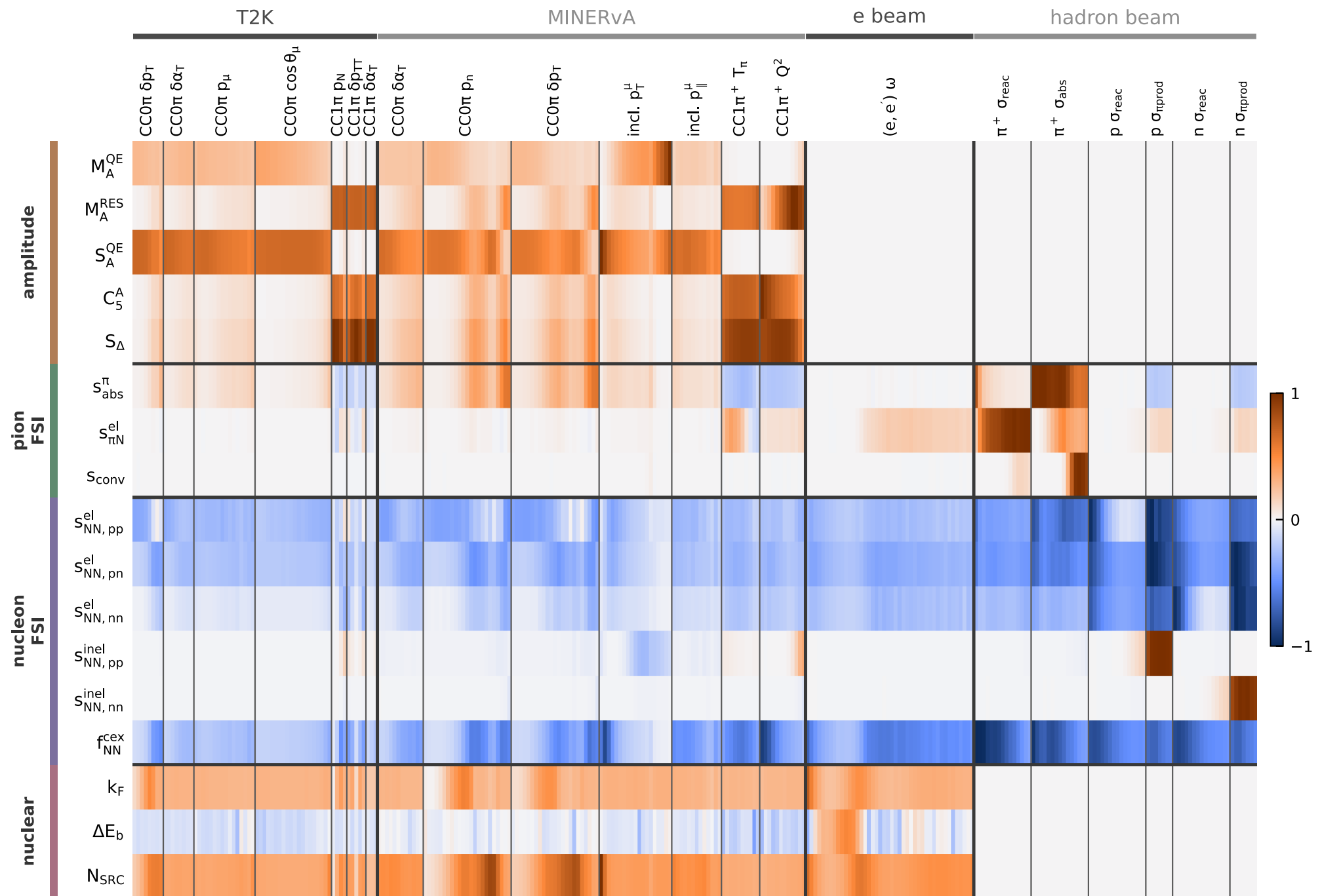
Example hadron beam on C and Ar



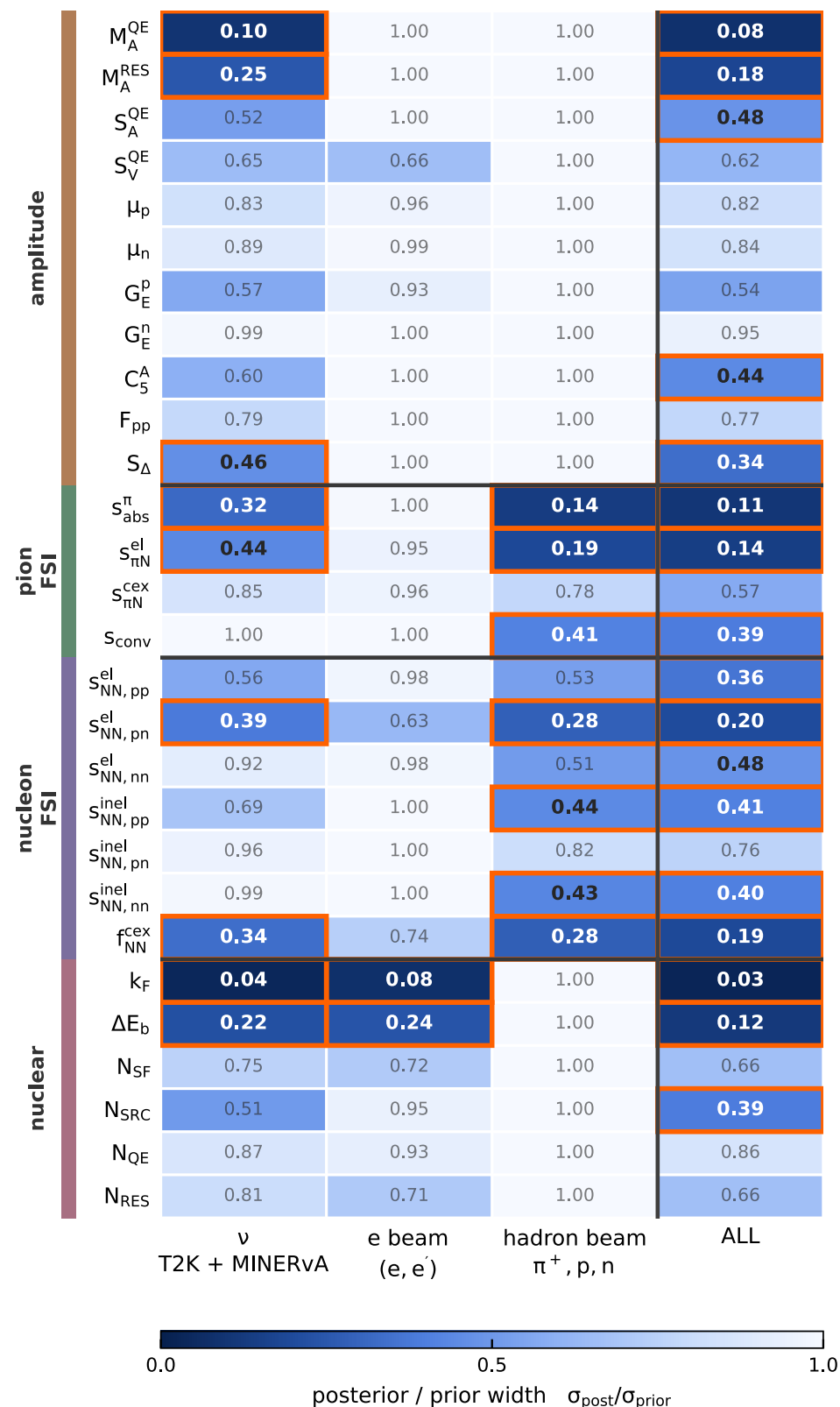
— Pre-fit (nominal) - - - Post-fit † Asimov data



Imagine we consider the following samples... all at once



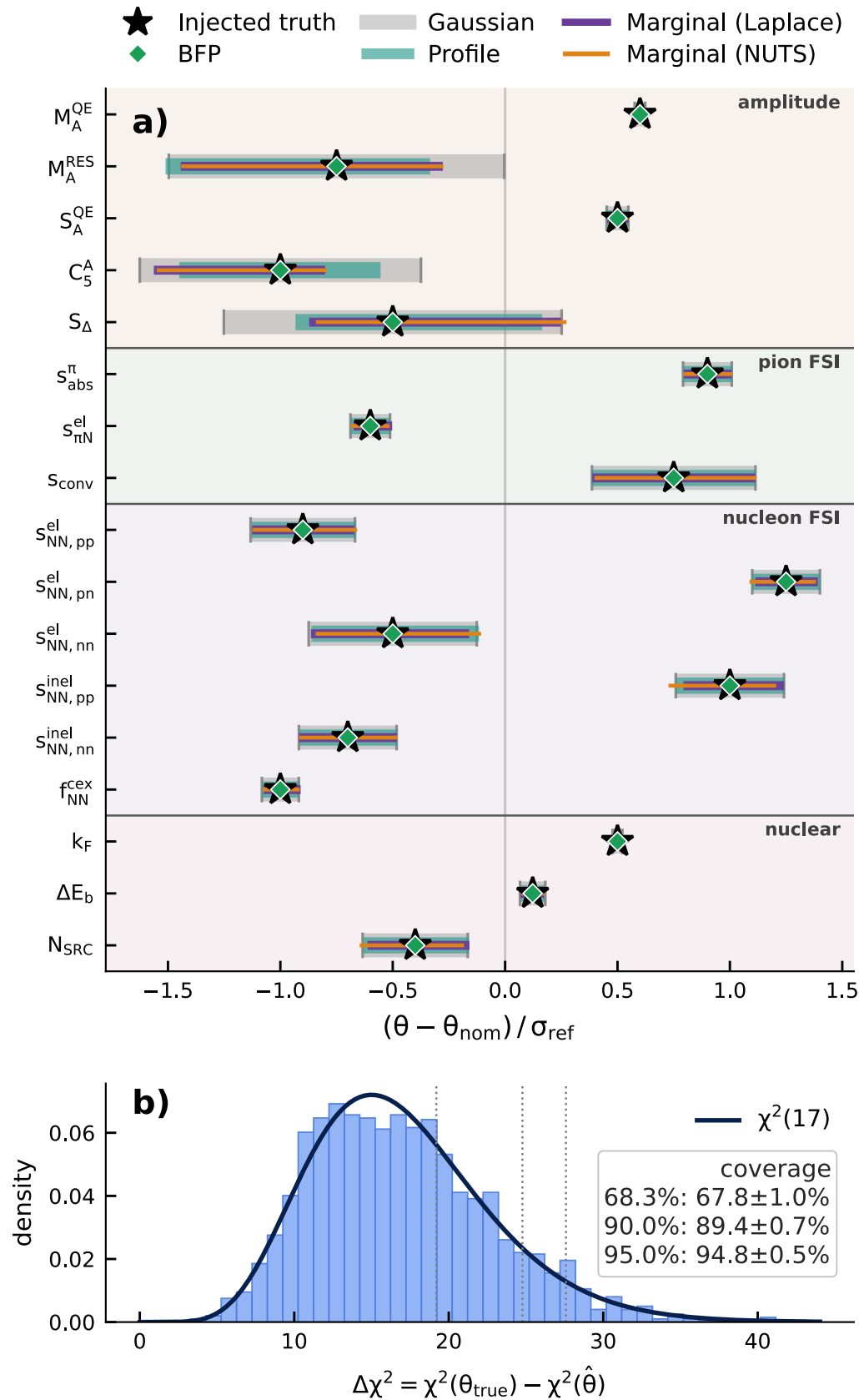
Are observables complementary? → Measurement design & design of experiments
 What bins contribute the most? → Measurement design & robustness studies



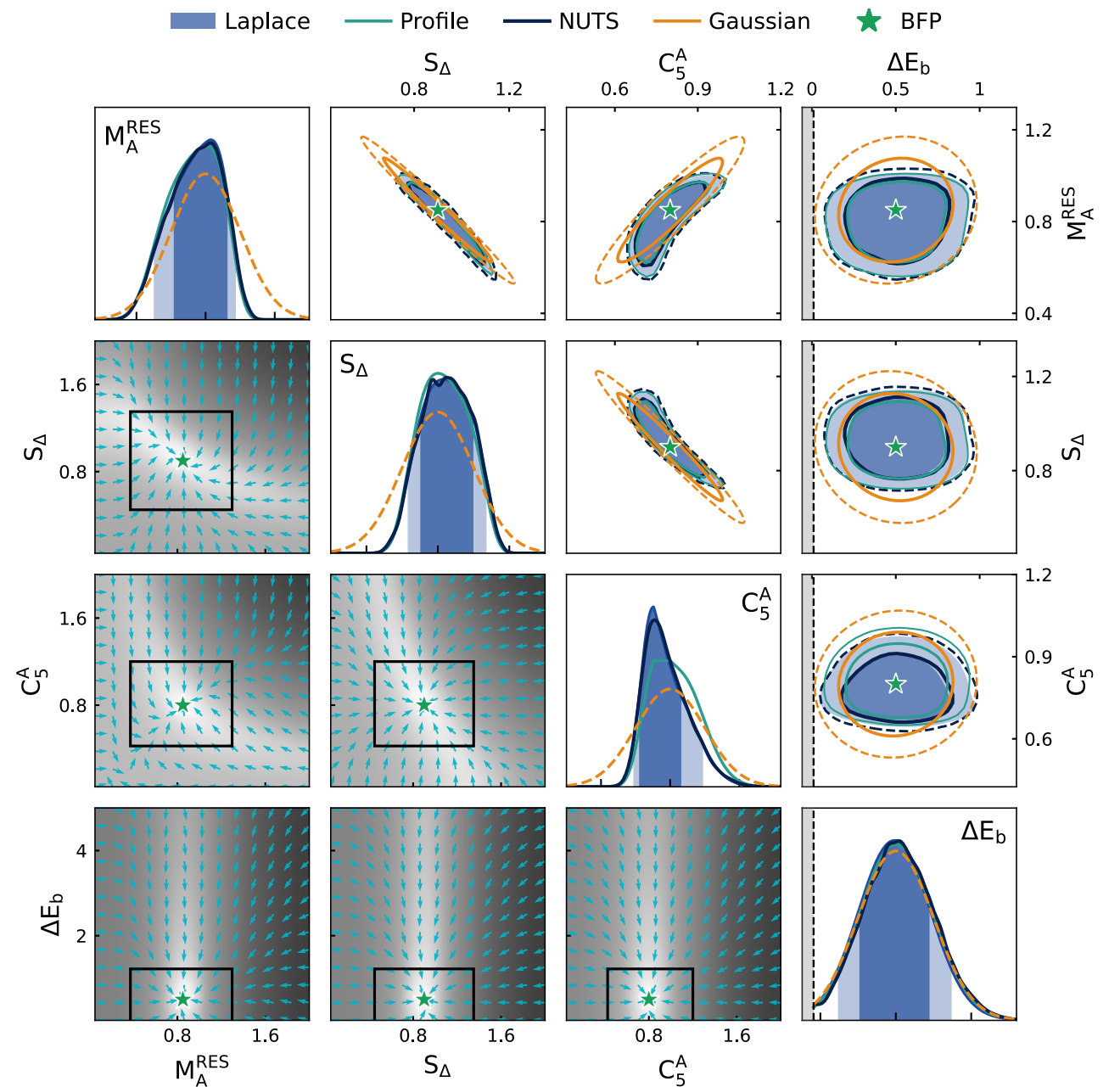
What about correlations?

If we define some arbitrary prior scale we can study what constraining power we have on model parameters.

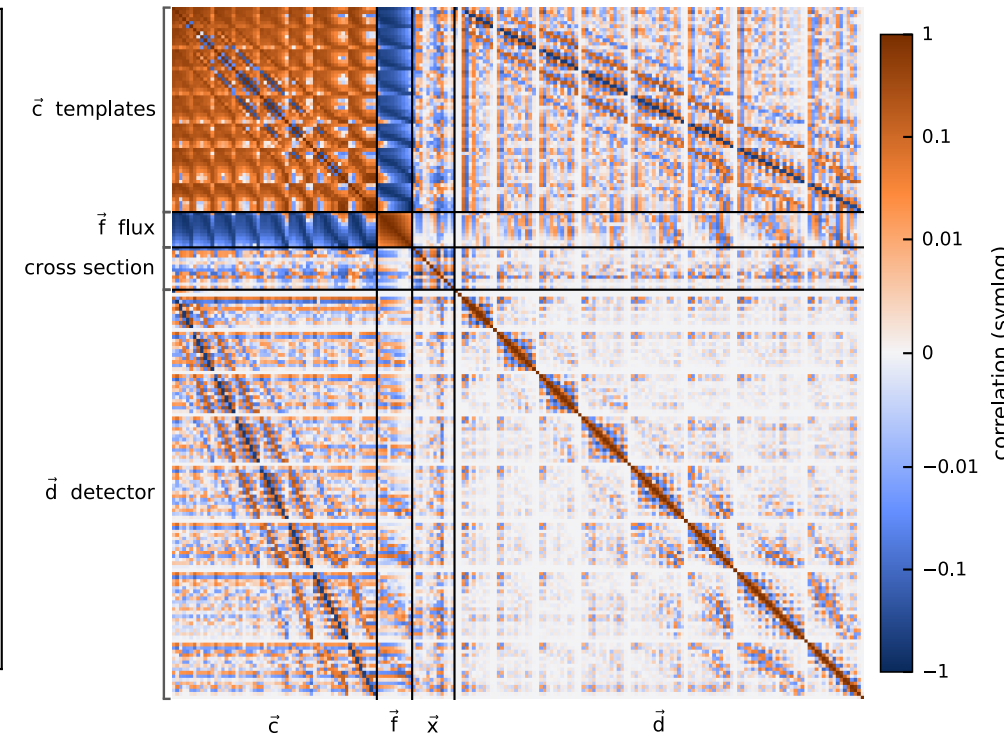
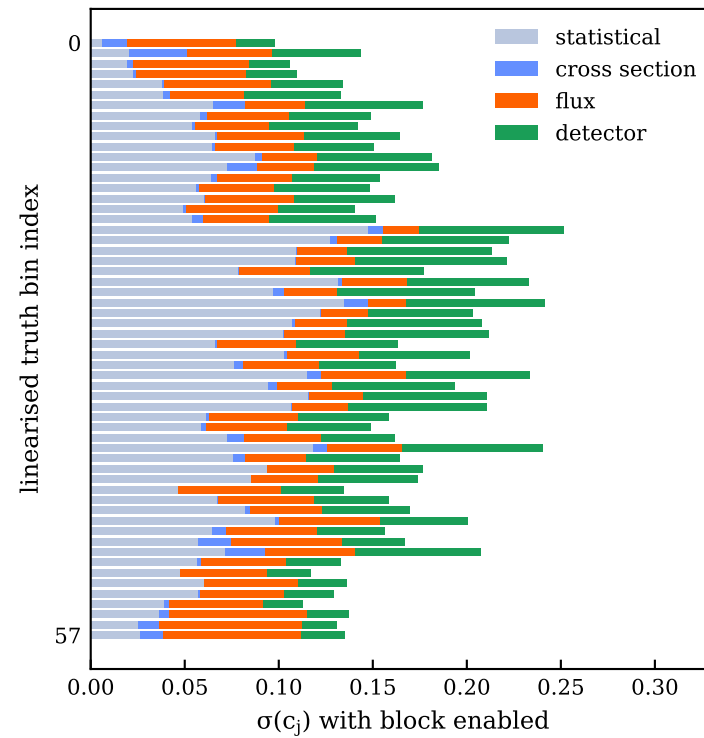
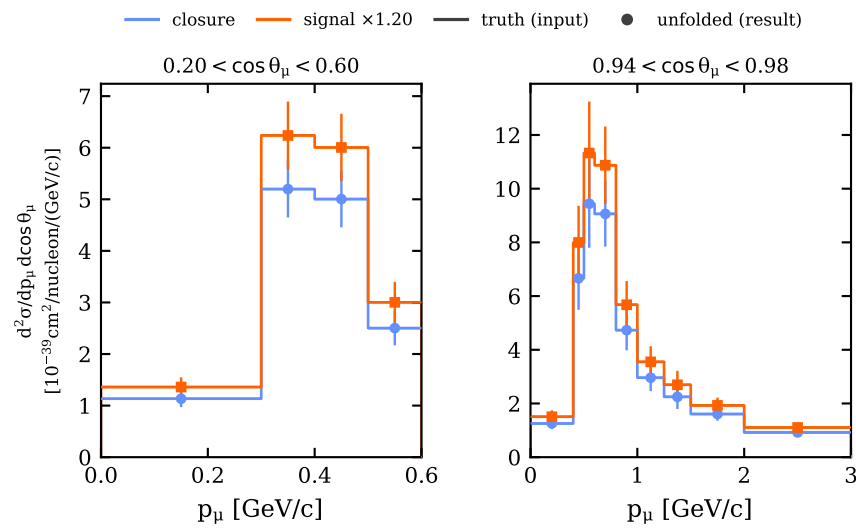
Allows to easily see what role have individual samples in joint fits and identify parameters that might benefit from samples of data different from those considered.



Any differentiable neutrino event generator is itself a “Fitter”!!!



And yes! it also works for unfolding after smearing into reconstructed space!



Conclusions